

The Economic Impacts of Hurricanes on Fisheries in the United States*

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Abstract

Hurricanes routinely traverse water and land along the Southeastern United States, bringing with them hazardous weather conditions at sea, which presumably hinder fishing operations. Yet, work examining the impact of hurricanes on fisheries has mostly focused on supply chain analyses and post-disaster recovery, without paying attention to the first link in a long chain of events: how hazardous weather affects fisheries production. Here, we quantify the economic toll of hurricanes on commercial fisheries in the Southeastern United States. We construct a novel, high-frequency county-level panel linking daily commercial landings and revenue to hurricane exposure from 2005 to 2022. Our results indicate that county-level wind exposure reduces daily commercial landings and revenue by up to 60 and 70% under major-hurricane-force winds. Monetizing responses indicate cumulative revenue losses of \$162.4 million over 2005–2022 due to hurricanes. These findings underscore the persistent vulnerability of coastal fisheries to extreme weather and provide an empirical foundation for targeted disaster preparedness and climate adaptation policy.

JEL: Q22, Q51, C54

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1 Introduction

Hurricanes pose a growing threat to coastal economies in the Southeastern United States. The frequency and intensity of North Atlantic hurricanes have increased since the 1980s, and major hurricanes are expected to become more common (Vecchi et al., 2021). Commercial fishing communities are particularly exposed, yet the precise economic toll of hurricane-force winds on fisheries production remains poorly quantified (Oxenford and Monnereau, 2018; Monnereau et al., 2021). This paper investigates the economic impacts of hurricanes on the commercial fishing sector in this region, which includes both the Gulf of America / Gulf of Mexico and the South Atlantic—two of the nation’s most productive and economically vital marine regions (Villaseñor-Derbez et al., 2026). Taking advantage of newly available data on hurricanes (Molina and Rudik, forthcoming), we develop a county-level daily panel of commercial landings and wind speed exposure that leverages variation in wind intensity from land-falling hurricanes. Our dataset contains all commercial landings in the Gulf and the South Atlantic for 293 species and all 31 hurricanes¹ that made landfall in the continental United States between 2005 and 2022. We use this high-resolution dataset to establish: (1) the causal effect of hurricane-wind exposure on commercial landings and revenue, and (2) the associated economic losses. Noting that counties experience varying wind intensities during the same event, our analysis shows that county-level daily landings and revenue decline sharply at the onset of exposure. Specifically, we find reductions of 8–11% under tropical-storm-force winds (39–73 mph), 35–41% under hurricane-force winds (74–110 mph), and 60–70% under major-hurricane-force winds (≥ 111 mph) within the 30-day after landfall.

Dynamic loss estimates show that losses begin shortly before exposure, remain statistically meaningful for about a week under tropical-storm-force winds, and persist for roughly two to four weeks under hurricane-force and major-hurricane-force wind exposure. Monetizing these dynamic effects implies cumulative revenue losses of about \$162.4 million over 2005–2022. Losses were especially large in severe seasons, reaching about \$71.5 million in 2020 alone, or roughly 13% of that year’s hurricane-season ex-vessel revenue.

Despite billions of dollars in federal disaster aid being disbursed to fishing communities over the last few decades (Bellquist et al., 2021; Deryugina, 2017) and a growing ecological literature

¹maximum sustained wind speeds *at the eye of the storm and over water* greater than 33 meters per second [m/s]

documenting hurricane impacts on marine ecosystems (Greening et al., 2006; Ramenzoni et al., 2020), studies that systematically translate these shocks into region-wide economic consequences for commercial fleets are scarce. This paper addresses that gap by combining three features absent from prior work, which include daily temporal resolution, a causal design, and an 18-year panel spanning the full Southeastern U.S. commercial fishing sector.

Existing economic assessments have largely focused on capital destruction and supply-chain bottlenecks following specific catastrophic storms, such as Hurricanes Katrina, Maria, and Ida (Petterson et al., 2006; Posadas, 2025; Agar et al., 2020). These post-disaster case studies provide vital insights into structural vulnerabilities, but they typically rely on aggregated ex-post survey data that bundles the immediate meteorological shock with prolonged logistical recovery. Moreover, the insights gleaned from these major storms are unlikely to be representative of less catastrophic but more frequent storms that often go unstudied. NOAA’s evaluation of Hurricanes Irma and Maria, for instance, reported \$106 million in revenue losses and \$117 million in damages across Florida, Puerto Rico, and the U.S. Virgin Islands (National Oceanic and Atmospheric Administration, 2018b,a,c), but those figures combine capital damage, forgone revenue, and downstream disruptions into a single estimate. For policymakers designing targeted adaptation strategies such as precisely timed financial relief, localized infrastructure investments, or species-specific interventions, this lack of resolution is a significant hurdle. Our design only isolates the dock-side losses in production and revenue, but does so over an 18-year panel that captures both anticipatory behavior and recovery dynamics of all hurricanes. Our estimates constitute then the lower bound of the economic toll in the industry, which also encompasses capital destruction and downstream spillovers that our production-focused design does not measure.

Beyond direct damage to infrastructure, existing research also recognizes the short-term behavioral and operational disruptions caused by the imminent threat of extreme weather. For instance, fishers often alter their spatial behavior or halt operations preemptively in response to hurricane warnings, which induces immediate losses in the days and weeks preceding landfall (Solís et al., 2013). Yet, because most existing economic impact assessments rely on monthly or annual data aggregates, the precise temporal dynamics of how a fishery responds and recovers remain obscured. Here, we address this gap by constructing a high-frequency, daily panel linking atmospheric wind exposure to county-level fisheries production. By systematically quantifying the non-linear, tempo-

ral shocks associated with hurricane exposure, we provide the high-resolution empirical foundation necessary to inform climate adaptation and disaster relief policies.

The paper proceeds as follows. Section 2 provides background information on fisheries in the region and documented impacts of hurricanes on fisheries. Section 3 describes the data we use in our analysis. Section 4 presents our methods and results. Section 5 concludes.

2 Background

2.1 Commercial Fisheries in the Southeastern United States

This study examines commercial fisheries across both the Gulf and South Atlantic regions, two of the nation’s most productive and economically vital marine regions. The Gulf region, extending from the southern tip of Florida westward along the coasts of Alabama, Mississippi, Louisiana, and Texas, generated approximately \$795 million in annual ex-vessel revenue in 2023, representing 15.7% of total U.S. revenues (National Marine Fisheries Service, 2026a). Its most iconic commercial fisheries include shrimp, snapper and groupers, menhaden, oysters, and blue crabs. As Gulf revenues move through the broader economy, they generate \$3.9 billion in sales, \$1.4 billion in income, and support about 53 thousand full- and part-time jobs (National Marine Fisheries Service, 2026b). The region’s processing and distribution sectors add further value, producing \$1.9 billion in value-added output; more than 2.4 times the value of unprocessed landings (National Marine Fisheries Service, 2026b).

The South Atlantic region, extending from North Carolina through the Florida Atlantic coast, generated approximately \$172 million in ex-vessel revenue in 2023, accounting for 3.4% of total U.S. revenues. Its commercial fisheries include shrimp, snapper–grouper complexes, dolphin and wahoo, and coastal migratory pelagic species. As South Atlantic revenues circulate through the regional economy, they support over 5,000 jobs, generate \$243 million in labor income, and contribute \$670 million in total sales (National Marine Fisheries Service, 2026b). Processing and distribution activities further amplify the region’s economic footprint, producing \$335 million in value-added output across coastal communities.

Despite their central role in sustaining coastal livelihoods and contributing to regional and national food security, fisheries in both the Gulf and South Atlantic are increasingly vulnerable

to hurricanes (Villaseñor-Derbez et al., 2026). These intense storms disrupt harvesting, damage vessels and infrastructure, impede processing and distribution, and generate substantial economic losses throughout the seafood supply chain. Quantifying the magnitude and persistence of these disruptions is therefore essential for understanding the resilience of commercial fisheries and the communities that depend on them.

2.2 Impacts of Hurricanes on Fisheries

Hurricanes affect fishing communities through multiple, simultaneous channels. These include operational disruption, capital destruction, and ecological degradation, whose combined economic toll has proven difficult to disentangle empirically. Ecological responses to weather can also be complex. Some studies find that elevated wind and storm activity temporarily raise catchability for certain species by altering their movement, including lobster and black sea bass (Jury et al., 1995; Wiernicki et al., 2020; Reustle et al., 2024). These effects, however, reflect routine turbulent conditions rather than the extreme, safety-limiting winds of a hurricane, under which fishing operations largely cease.

Most existing economic evidence comes from post-disaster assessments of individual catastrophic storms. Studies of Hurricane Katrina, for example, documented widespread destruction of vessels, docks, and processing facilities across the Gulf coast, with recovery timelines shaped as much by logistical bottlenecks such as power outages, fuel shortages, or labor displacement, as by the initial storm damage itself (Pettersen et al., 2006; Impact Assessment Inc, 2007; Ingles, 2007). More recent assessments of Hurricanes Irma, Maria, and Ida have reinforced these findings, reporting losses that span damaged capital, forgone revenue, and downstream supply-chain disruptions (Agar et al., 2020; Posadas, 2025; Caffey et al., 2022).

A common limitation of these assessments is that they bundle multiple margins of loss into aggregate damage figures. NOAA’s evaluation of Hurricanes Irma and Maria, for instance, reported \$106 million in overall revenue losses and \$117 million in physical damages across Florida, Puerto Rico, and the U.S. Virgin Islands (National Oceanic and Atmospheric Administration, 2018b,a,c), while a Louisiana Sea Grant assessment of Hurricanes Laura, Delta, Zeta, and Ida estimated roughly \$579 million in total fisheries losses (Caffey et al., 2022). Because these figures often combine capital destruction, forgone harvesting revenue, and broader economic spillovers, they offer limited

guidance for designing targeted interventions such as precisely timed financial relief or localized infrastructure investments. Furthermore, because each assessment includes different sectors of the economy, it is difficult to compare economic damages across storms and assessments.

A smaller literature has attempted to isolate production-side effects directly. Solís et al. (2013) used a stochastic production frontier to estimate the bi-weekly effect of hurricanes on the Gulf of America/Gulf of Mexico commercial grouper fishery over 2005–2009. They found fleet-wide immediate revenue losses of at least \$927,000 over the period, with regional production declines of 50–60% in the hardest-hit areas during peak hurricane seasons. Notably, in several cases, such as Hurricanes Gustav and Ike, fishing ceased entirely in affected regions, and losses could not be estimated within their framework. Their study represents one of the few efforts to separate the operational shock from capital damage, but its focus on a single species group, short time horizon, and bi-weekly frequency limits the ability to trace the precise timing and cross-fishery scope of the disruption.

The precision of these estimates matters directly for policy. Under the Fishery Resource Disasters Improvement Act, state governors may request a federal fishery disaster determination from the Secretary of Commerce, and NOAA evaluates such requests using sector-specific revenue losses. Revenue losses above 80% support a positive disaster determination and those between 35% and 80% are subject to further review (NOAA Fisheries, 2025). Once approved, federal relief funds are administered by NOAA Fisheries and supplemented by FEMA programs including Small Business Administration Disaster Loans and Disaster Unemployment Assistance (FEMA, 2025). At the state level, agencies such as the Louisiana Department of Wildlife and Fisheries and the Florida Fish and Wildlife Conservation Commission conduct their own damage assessments and distribute additional assistance. Yet survey evidence consistently finds that formal aid arrives slowly, and fishers rely heavily on informal community networks like family, friends, and local organizations for immediate post-storm recovery (Agar et al., 2020; Dunning, 2020; Ingles, 2007). The gap between the aggregate damage figures used to trigger relief and the actual temporal pattern of production losses underscores the need for higher-resolution causal estimates.

We build on this body of work by expanding our analysis to cover the full Southeastern U.S. commercial fishing sector, extending the panel to 18 years, and moving to daily resolution. This approach allows us to capture both anticipatory reductions in fisheries production before exposure

and the gradual recovery path afterward.

3 Data

This section describes the construction of the panel dataset used in our analysis. We combine detailed commercial fisheries logbook records with high-frequency hurricane exposure data to quantify their effects on fisheries production across the Southeastern United States. We document the data sources, processing steps, and summary statistics, and provide details on how we link hurricane events to county-level fishing outcomes by species group and date.

3.1 Fisheries Data

Our primary fisheries data are sourced from the SEDAT Commercial Landings Extraction, a standardized database managed by the Southeast Fisheries Science Center. This dataset integrates high-resolution, trip-level records from the National Marine Fisheries Service Southeast Coastal Fisheries (NMFS) Logbook Program with comprehensive landings data sourced from the Atlantic Coastal Cooperative Statistics Program and the Gulf Fisheries Information Network.

Each record provides information on departure and offloading dates, fishing location, species-specific landings, revenue, fishing gear, crew size, days away, and landing site identified at the county level via Federal Information Processing System (FIPS) codes. NMFS acknowledges that the SEDAT dataset contains a known limitation for shrimp species (e.g., white, brown, and pink shrimp). We note that between 2005 and 2015, redundancies in the Gulf Shrimp System resulted in some duplicate port agent and trip ticket records for the same trip, which inflate reported landings totals.² In addition, about 2.9% of trip records show a landed weight of -1 , which we assume are missing entries and are therefore removed.

We process these data through several stages to ensure consistency and analytical usefulness. First, we clean and standardize the raw fisheries records, mapping individual species into seven broader functional groups for analysis: crustaceans, mollusks, reef fish, inshore fish, pelagic fish, sharks and rays, and other species. We then aggregate the data to construct a daily panel of

²These duplicate records inflate the *level* of reported crustacean landings but are unlikely to bias our estimates, which are identified from within-event variation between the pre- and post-exposure windows. Duplication would bias the estimated effect only if its incidence were correlated with the timing of hurricane exposure, which there is no reason to expect. Appendix B.5 shows this is not the case.

1,778,313 observations at the county $\times$ date $\times$ species-group level. In constructing the daily panel, each trip’s landings and revenue are assigned to its landing date (i.e., the date at which catch enters the market). For each cell, the panel reports landings and revenue, as well as the number of trips, the number of dealers, total effort, the most common gear type, the number of gear types used, the most common fishing area, and the number of fishing areas represented.

Table 1 reports summary statistics for the fisheries data for days with recorded landings. The table shows substantial heterogeneity across species groups in both scale and seasonality. Pelagic fish dominate in terms of physical volume, with mean annual landings of 355 million kilograms and mean landings of 37 thousand kilograms on active fishing days. Crustaceans, by contrast, dominate in value, generating \$523 million in mean annual revenue, far exceeding pelagic fish (\$104 million), mollusks (\$77 million), and reef fish (\$70 million). Crustaceans also account for 170 million kilograms in mean annual landings, making them the largest non-pelagic fishery by volume. The remaining groups are considerably smaller in commercial scale, with other species generating \$39 million annually, inshore fish \$19 million, and sharks and rays \$3 million, respectively.

Table 1: Summary Statistics by Species Group

Species Group	Daily		Annual		
	Mean Landings (SD)	Mean Revenue (SD)	Mean Landings (SD)	Mean Revenue (SD)	Mean Days per Year (SD)
Crustaceans	6,059 (13,751)	18,698 (54,361)	169.5M (20.5M)	523.1M (160.8M)	231 (134.3)
Pelagic Fish	37,376 (277,198)	10,959 (60,460)	354.8M (58.3M)	104.0M (23.9M)	109 (109.6)
Mollusks	2,328 (7,682)	9,427 (21,958)	19.0M (7.1M)	77.1M (26.0M)	152 (124.2)
Reef Fish	865 (1,937)	5,651 (14,686)	10.6M (894.5K)	69.6M (14.3M)	141 (122.4)
Inshore Fish	676 (3,217)	1,175 (4,893)	11.0M (2.4M)	19.1M (3.2M)	155 (126.4)
Sharks & Rays	944 (3,954)	1,449 (3,134)	1.9M (775.1K)	2.9M (1.1M)	46 (62.0)
Other	865 (3,595)	1,827 (7,489)	18.4M (3.3M)	39.0M (4.3M)	148 (121.6)

Note:

The table shows summary statistics for each species group excluding days with no landings. Daily landings and revenue are measured in kilograms (kg) and US dollars (\$), respectively. Annual values represent the average across all years in the dataset. Average days per year indicates the average number of days per year that a typical county lands each species group, providing a measure of seasonality (lower values indicate more seasonal fishing patterns). Standard deviations are reported in parentheses.

The data also reveal marked differences in seasonality, measured by the average number of days per year that a typical county records landings in each group. Crustaceans are landed most consistently throughout the year, with an average of 231 days per county-year. Mollusks, reef fish, inshore fish, and other species are also relatively regular fisheries, with roughly 141 to 155 landing days per year. By contrast, pelagic fish are more episodic, averaging 109 landing days per county-year, while sharks and rays are the most seasonal group, with only 46 days per year on average.

Appendix Tables A1 and A2 provide species-level detail on the concentration of value and volume within the fishery. The revenue rankings show that economic value is concentrated in a relatively small number of species, led by northern white shrimp, northern brown shrimp, blue crab, eastern oyster, and menhadens. The landings rankings are even more concentrated in high-volume species, led overwhelmingly by menhadens, followed by bait shrimp, northern white shrimp, blue crab, and northern brown shrimp.

3.2 Hurricane Data

Hurricane exposure data are sourced from Molina and Rudik (forthcoming) and include both observed and forecast measures of wind speed (m/s) and precipitation (mm), as these are two of the main drivers of damage. The dataset encompasses all 31 Category 1 or higher hurricanes that made landfall in the continental United States between 2005 and 2022 and contains 74,276 county-by-day records of maximum wind speed and total precipitation. Forecast data are available in 1- 2- and 3-day leads, depending on the storm.

Table 2 summarizes the exposure and forecast performance for both metrics. Average observed and forecast wind speeds are 7.25 m/s and 6.61 m/s, respectively. These relatively low means reflect the geographic breadth of the data, which includes many county-level observations located outside the high-intensity path of the storm center; Maximum observed and forecast wind speeds are considerably higher, at 67.15 and 35.84 m/s, respectively. The data highlights a significant discrepancy in variability: the 3-day forecast shows substantially lower variability ($SD = 6.41$ m/s) than the observed values ($SD = 8.51$ m/s). This gap is largely attributable to rapid intensification events, which consist of abrupt increases in storm strength that current meteorological models often fail to predict accurately.

Federal emergency responses are triggered by a storm’s projected wind strength *over land*. There are three exposure levels that are the most relevant: tropical depressions, with sustained winds of 10–17.5 m/s (6–10 mph), tropical storms, with sustained winds of 17.5–32.9 m/s (39–73 mph), and hurricanes, which fall into Categories 1–5 and begin at 33.0 m/s (74 mph). Hurricanes are classified using the Saffir-Simpson Hurricane Wind Scale which measures their wind intensity *over water*. Hurricanes can be further divided into hurricanes (categories 1–2) and major hurricanes (categories 3–5). The transition from tropical storm-force to hurricane-force winds serve as the

Table 2: Wind and Precipitation Exposure During Hurricane Season

Variable	Mean	SD	Min	Max
Observed Wind (m/s)	7.25	8.51	0.10	67.15
Average 3-Day Forecast Wind (m/s)	6.61	6.41	0.21	34.84
Wind Forecast Absolute Error (m/s)	2.02	3.71	0.00	35.07
Observed Precipitation (mm)	19.34	43.79	0.00	488.65
Average 3-Day Forecast Precipitation (mm)	14.68	26.93	0.00	250.09
Precipitation Forecast Absolute Error (mm)	14.18	32.12	0.00	329.65
Counties Affected per Season (Tropical Storm+)	75.92	38.89	34.00	172.00
Counties Affected per Season (Hurricane+)	11.25	10.77	0.00	34.00

Note:

Summary statistics of observed and average 3-day forecast maximum sustained wind speeds (m/s) and precipitation (mm) during hurricane season (June-November) across all counties. Forecast absolute errors are calculated as the absolute difference between observed and forecast values. Counties affected per season shows the number of counties experiencing tropical storm conditions (17.5 m/s) or hurricane conditions (33.0 m/s) or higher during each hurricane season.

primary signal for pre-landfall adaptive actions such as warning and mobilization of emergency resources (National Weather Service, 2024). In our data, tropical-storm-force exposure occurs about seven times as often as hurricane-force exposure.

Precipitation data exhibits even higher levels of uncertainty than wind, with a mean absolute error nearly equivalent to its forecast mean (Table 2). This underscores that while wind speed is the primary coordinating signal for policy, precipitation remains significantly more difficult to predict and less integrated into early-stage institutional response. While our main analysis will focus on wind, our results are robust to controlling for precipitation (Appendix B.1).

Finally, the localized nature of these impacts is evident. On average, 76 counties per season experience tropical storm-force winds, but only 11 counties experience the hurricane-force winds that drive the most severe economic damages.

Our exposure measure captures wind speed at the county where landings are recorded rather than at the vessel’s fishing grounds. Winds are typically stronger and more variable at sea than on land, and their relevance to a given trip diminishes as fishing activity moves farther offshore. We nonetheless rely on county-level land exposure for three reasons. First, the operational margins

that drive our results—whether to leave port, when to return, and whether shoreside infrastructure (i.e., docks, ice, fuel, processing) is functional—are governed by conditions on land, not at the fishing grounds. Second, because vessels traverse multiple, often unobserved offshore areas within a single trip, constructing a singular at-sea exposure metric would be highly sensitive to arbitrary assumptions. Third, county-level exposure provides a consistently observed metric for every landing record, whereas attempting to reconstruct trip-specific offshore conditions would introduce systematic data gaps. County land exposure should therefore be read as a proxy for the storm conditions that disrupt a county’s fishing operations, and likely a lower bound on the winds actually encountered at sea.

3.3 Fisheries and Hurricanes

To assess hurricane effects on fisheries production, we link daily county-level storm-force wind exposure to daily fisheries production by species group. The resulting panel combines fisheries and hurricane data at the county $\times$ date $\times$ species-group level, yielding a balanced daily dataset with 12.7 million observations for 276 counties in the Southeastern United States over 2005–2022. Each observation includes landings (kg), revenue (US\$), number of trips, and primary gear type. The panel also includes hurricane exposure measures based on observed wind speed and precipitation, forecast wind and precipitation at 1-, 2-, and 3-day lead times, average forecast exposure across lead times, and corresponding forecast errors. Unlike previous work that used data on just a few, mostly catastrophic storms, this novel dataset contains information on a wider range of storm events, thus allowing us to perform more nuanced and detailed analyses on the effects of hurricanes on fisheries production across the Southeastern United States (See Table A3).

Figure 1 presents the time series of average monthly landings per county over the study period. Circular markers indicate landfall months. The figure shows substantial variation in fisheries production, with average monthly landings per county ranging from 34.3 to 520.2 tons (median: 128.0 tons; mean: 176.7 tons). Across the 26 exposure months, average monthly landings per county were higher than in non-exposure months (231.4 tons versus 169.2 tons), although this descriptive difference likely reflects both seasonality in fishing activity and the timing of storm exposure. More broadly, the figure suggests some pronounced seasonality but no obvious long-run trend in fisheries production over the sample period. Marker size reflects the number of counties affected on the

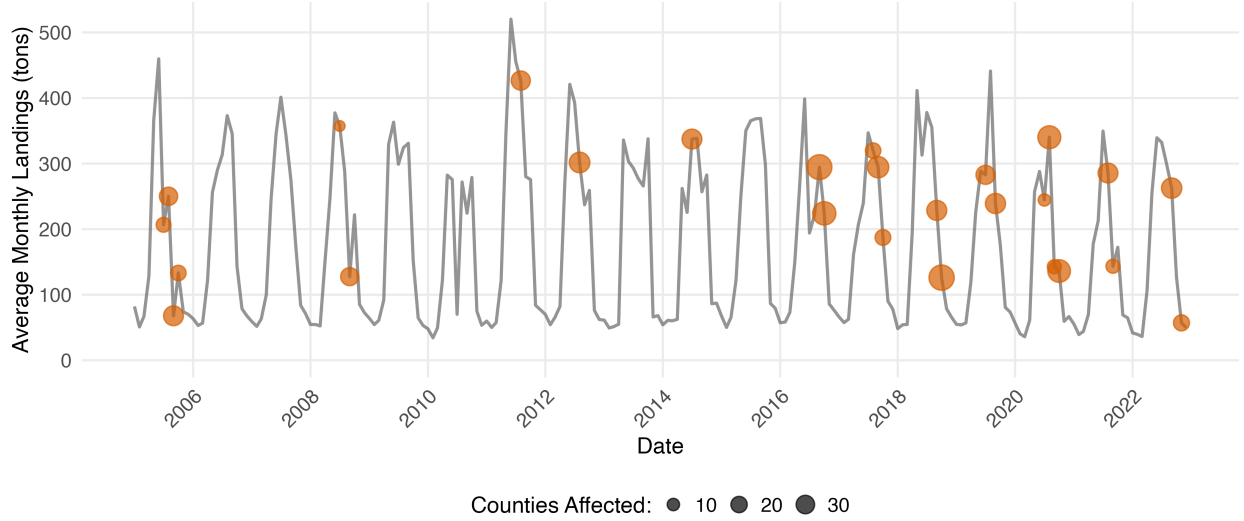


Figure 1: Average monthly landings and hurricane exposure per county. Markers indicate months in which a hurricane made landfall in the continental United States. Marker size reflects the number of counties affected on the strongest exposed day in that month.

strongest exposed day in each month, with larger exposure episodes appearing more frequently after 2016.

4 Methods and Results

4.1 Static Effects

We begin by estimating the immediate effect of county-level exposure to storm-force winds on commercial fisheries using a pre-post fixed-effects panel regression framework. Our primary specification takes the form:

$$\text{asinh}(Y_{ist}) = \alpha_{im} + \beta_k \text{Post}_{ist} \times \text{ExposureLev}_{k,ist} + \gamma_y + \delta_d + \lambda_s + \epsilon_{ist}. \quad (1)$$

Here, Y_{ist} represents our outcome variables (landings or revenue) for county i , species group s , and date t . Post_{ist} is a binary indicator equal to 1 for observations in the 30-day post-exposure period and 0 for observations in the 30-day pre-exposure period, where exposure periods are defined around county-level events. Temporally, nearby treated days for a given county-species unit are grouped into a single exposure event to avoid treating closely spaced exposure days as separate shocks. The estimation sample is then constructed from the 30 days before and 30 days after that

event following the recovery patterns in the dynamic estimation below. Appendix B.3 shows our results are not sensitive to the window choice.

ExposureLev_{*k,ist*} is a categorical variable, indexed by *k*, indicating the maximum wind speed reached during the county-level exposure event. The potential *k*-levels are: Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), or Major Hurricane (≥ 50 m/s). It is worth noting that these exposure levels may occur within a single hurricane episode. For example, a county directly under the eye wall of a category 5 hurricane will experience far more destructive winds than a county only exposed to the outskirts of the same hurricane. This differentiation is important because, as shown in Table 2, realized local wind exposure varies substantially across counties within the same hurricane. By classifying observations according to county-level wind intensity, we can estimate whether fishing outcomes respond differently to weaker or more severe local wind exposures rather than assuming all hurricane events are homogeneous. Appendix A.1 reports the distribution of exposure events and pre/post observations across the four categories.

The interaction Post_{*ist*} $\times$ ExposureLev_{*k,ist*} captures the differential effect of each exposure level in the post-exposure period relative to the pre-exposure period, with the pre period serving as the reference. The coefficients β_k thus represent the change in outcomes between the pre-exposure and post-exposure windows for each exposure level. The outcome is transformed using the inverse hyperbolic sine (asinh) function to handle skewed distributions and accommodate zero values. Appendix B.6 shows that these magnitudes are robust to functional form.

We control for several sources of variation through fixed effects. α_{im} represents county-by-month-of-year fixed effects that capture time-invariant county characteristics and seasonal patterns in fishing activity. γ_y represents year fixed effects that control for annual trends in fisheries production and broader economic or environmental conditions. δ_d represents day-of-week fixed effects that account for weekly patterns in fishing behavior. λ_s represents species group fixed effects that account for species-specific factors affecting catchability, market conditions, and fishing techniques.³

³We do not directly control for stock size, which is unavailable at daily frequency and not estimated for all species groups. Due to their complex methodologies, comprehensive stock evaluations are usually updated on a five- to seven-year cycle. Instead, most of the year-to-year variation in stock size is absorbed by our county-by-month-of-year, year, and species-group fixed effects, which jointly capture seasonal abundance, annual stock conditions, and species-specific catchability. Residual bias would require stock size to shift discontinuously within the narrow pre/post window, which is unlikely over the 30-day horizon we study.

We address spatial dependence by computing Conley standard errors following Conley (1999).

Figure 2 presents the results. The estimates indicate substantial declines in fisheries production in the post-exposure period, with effects that increase with exposure intensity. Interpreting the coefficient magnitudes as approximate semi-elasticities⁴, tropical-depression exposure is associated with no meaningful post-exposure change in landings or revenue in the 30-day specification. While exposure to tropical depression is statistically indistinguishable from zero, tropical-storm exposure is associated with declines of about 8% in landings and 11% in revenue. Hurricane exposure is linked to declines of roughly 35% in landings and 41% in revenue, and major hurricane exposure is associated with declines of about 64% and 71%, respectively.

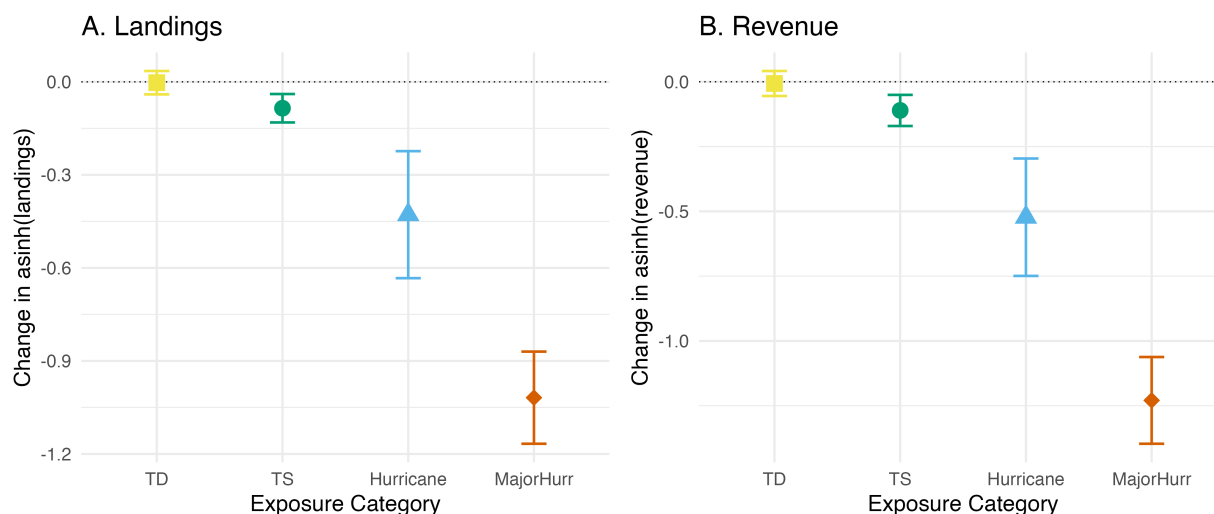


Figure 2: Pre-post effects of hurricane exposure on commercial fisheries by exposure level. Panel A shows the estimated change in the inverse hyperbolic sine of daily landings, while Panel B shows the corresponding change in the inverse hyperbolic sine of daily revenue. Each coefficient represents the difference between the 30-day post-exposure and 30-day pre-exposure windows around a county-level exposure event for the indicated wind exposure level. Levels are Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s; Categories 1–2), and Major Hurricane (≥ 50 m/s; Category 3 or higher). Bars represent 95% confidence intervals based on Conley standard errors. The regressions include county-by-month, year, day-of-week, and species-group fixed effects. The estimation sample contains 837,655 observations.

The gradient of effects across exposure levels illustrates that hurricane impacts scale non-linearly with wind intensity. These estimates suggest that major-hurricane wind exposure generates more

⁴Because the response variables are transformed using the inverse hyperbolic sine function, these magnitudes should not be interpreted as exact percentage effects (Bellemare and Wichman, 2020). For moderate and large values of the outcome, the transformation behaves similarly to a logarithm, so $\exp(\beta) - 1$ provides a useful approximation. This approximation is less precise near zero, so the reported percentages should be read as descriptive guides rather than exact semi-elasticities.

than four times the economic losses associated with tropical-storm-level winds. This pattern suggests that while moderate wind exposure temporarily disrupts operations, severe hurricane exposure triggers substantially larger and more persistent economic impacts. These amplified impacts likely arise from widespread infrastructure damage, prolonged safety restrictions, and cascading operational constraints that suppress fisheries activity far beyond a few days of rough seas.

These results are also robust to controlling flexibly for contemporaneous precipitation (Appendix B.1). When the estimates are disaggregated by species group, the non-linear response appears broadly across the fishery but is concentrated in a subset of groups: crustaceans are by far the most severely affected, followed by inshore fish and other species, while mollusks, pelagic fish, and reef fish exhibit clear but more moderate declines (Appendix B.2). Re-estimating the model in levels yields a nearly identical pattern (Appendix B.4).

We can also decompose the hurricane-wind effect into an expected component—driven by forecast information available to fishers before exposure, and a surprise component arising from forecast errors (Appendix D). The expected channel accounts for roughly three-quarters of the total effect, but forecast errors generate significant additional losses under hurricane-force and major-hurricane-force exposure (Appendix Table D2), consistent with rapid intensification events leaving fishers and infrastructure insufficiently prepared.

While this pre-post specification shows that hurricane exposure is associated with large average declines in production, it compresses the entire adjustment path into a single coefficient. In particular, the 30-day average blends any anticipatory reductions before exposure with the contemporaneous shock and subsequent recovery, making it impossible to determine how quickly, or whether, fisheries production returns to its baseline. We next estimate a dynamic model that traces the daily trajectory of disruption and recovery for each exposure level.

4.2 Dynamic Effects

Exposure events are constructed exactly as in the static analysis. For each county $\times$ species-group unit, hazardous-wind days occurring within 60 days—twice the 30-day estimation window—are grouped into a *single episode*, which we classify by the maximum county-level wind intensity reached during that episode. That is, the intensity actually experienced in that county rather than any given storm’s headline category. The exposure windows of distinct events cannot overlap, and every

observation is assigned to at most one event.

We estimate the following event-study specification:

$$\text{asinh}(Y_{ist}) = \alpha_{im} + \gamma_y + \delta_d + \lambda_s + \sum_{\tau \neq -7} \beta_{\tau k} \mathbf{1}\{\text{RelTime}_{ist} = \tau\} \times \text{ExposureLev}_{k,ist} + \epsilon_{ist}. \quad (2)$$

Here, Y_{ist} denotes landings or revenue for county i , species group s , and day t . RelTime_{ist} indexes the number of days relative to the exposure event, with day 0 corresponding to the event date. We estimate effects for each day in a symmetric 30-day window around the event, omitting day -7 as the reference period. The coefficients $\beta_{\tau k}$ therefore trace the evolution of outcomes before, during, and after exposure for each wind-intensity category. As in the pre-post specification, α_{im} denotes county-by-month fixed effects, γ_y year fixed effects, δ_d day-of-week fixed effects, and λ_s species-group fixed effects. Standard errors are computed using the same Conley correction as in the static analysis.

Figure 3 presents the results. They show that fisheries production contracts before exposure, and the onset of this anticipatory decline scales with storm severity. The pattern begins about one day before landfall for tropical-depression exposure, three days for tropical-storm- and hurricane-force exposure, and five days for major-hurricane exposure. This is consistent with fishers reducing activity in advance of hazardous conditions, but also extending operations to the very last minute. Appendix C.1 presents these dynamics separately by exposure category.

Second, and as expected, the largest contemporaneous and short-run losses occur on impact and in the days immediately afterward. These losses increase sharply with exposure intensity. Using the coefficients as rough semi-elasticity proxies, day-0 effects are small and statistically imprecise for tropical depressions, but imply declines in landings of about 12%, 32%, and 67% for tropical storms, hurricanes, and major hurricanes, respectively. The corresponding day-0 declines in revenue are about 15%, 38%, and 73%.

Third, recovery slows markedly as exposure intensifies. Tropical-depression effects are short-lived and become statistically indistinguishable from zero within a few days after exposure. Tropical-storm effects remain statistically significant through day 14, but are close to zero by day 30. Hurricane effects persist much longer, remaining statistically significant through day 29, although by day 30 the point estimates imply smaller and statistically imprecise declines of about 15% in landings

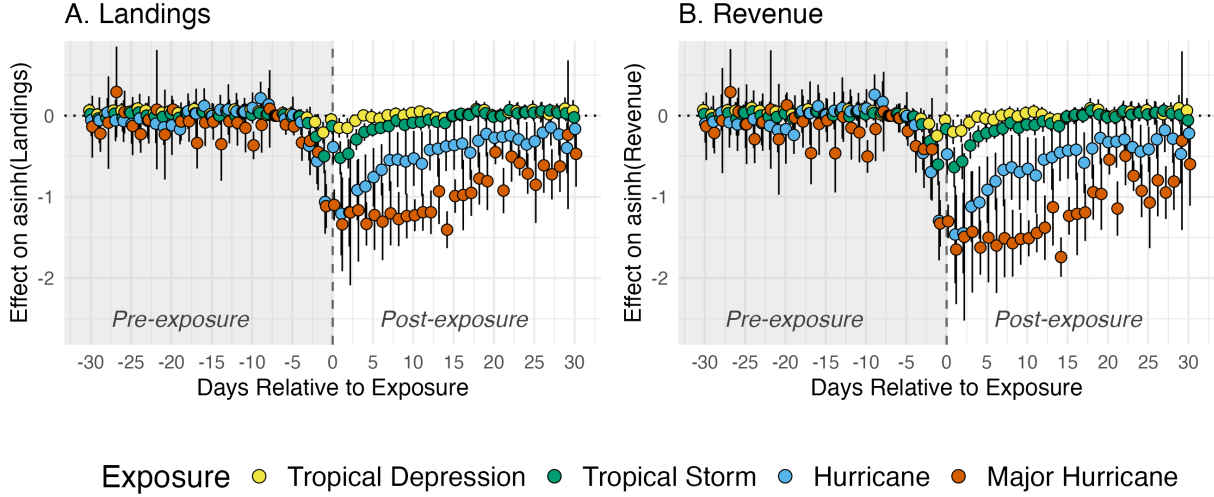


Figure 3: Dynamic effects of hurricane exposure by storm category. Panel A shows the estimated effect on the inverse hyperbolic sine of daily landings, and Panel B shows the estimated effect on the inverse hyperbolic sine of daily revenue), for each day relative to county-level exposure (day 0), disaggregated by exposure level: Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), and Major Hurricane (≥ 50 m/s). Points represent coefficient estimates, and bars denote 95% confidence intervals. The regressions include county-by-month, year, day-of-week, and species-group fixed effects. The estimation sample contains 837,655 observations.

and 20% in revenue. Major-hurricane effects are the most persistent, with coefficients at the end of the window implying declines of about 37% in landings and 45% in revenue.

A natural concern is whether the pre-landfall declines reflect a behavioral response to the impending storm or a spurious correlation. We argue that forecast-driven anticipation is part of the causal effect of the storm. Fishers who observe forecasts curtail operations before hazardous conditions arrive, whereas a spurious correlation would appear as systematic divergence prior to exposure unrelated to the storm. A couple features indicate the former is taking place: i) The leads preceding the onset of anticipation are statistically insignificant, showing no differential pre-trend. ii) The onset of the decline scales with storm severity, mirroring forecast products whose watches and warnings are issued 36–48 hours in advance and whose effective planning horizon lengthens for stronger, better-forecast storms. Our forecast decomposition (Appendix D) further clarifies that roughly three-quarters of the total wind effect comes from the expected channel, which is consistent with a response driven by anticipation of the storm rather than a pre-existing trend.

Overall, the dynamic evidence complements the pre-post results. Fisheries production decreases

in advance of storm arrival, exhibits sharp contemporaneous disruptions around exposure, and recover gradually afterward. Translating these coefficients into cumulative revenue losses, Figure 4 summarizes the economic magnitude of these disruptions by county-level exposure intensity. Aggregate losses are concentrated in counties exposed to hurricane-force and tropical-storm-force winds, with the largest total revenue losses arising from hurricane-force exposure, followed by tropical-storm-force exposure, major-hurricane-force exposure, and tropical-depression-force exposure. Note that there is substantial heterogeneity across space, with cumulative losses concentrated in a small number of highly exposed coastal states. Appendix C.2 shows that these dynamic responses are broadly present across species groups, although magnitudes vary across fisheries. Appendix C.3 replicates the dynamic analysis in levels and shows qualitatively similar patterns.

The cumulative revenue estimates show that losses are shaped not only by local wind intensity but also by the duration and spatial extent of disruption. Summing the day-specific event-study effects over the estimated loss windows implies total net revenue losses of about \$162.4 million over 2005–2022 (in 2022 US\$).⁵ Of this total, counties exposed to hurricane-force winds account for roughly \$95.5 million (59%), counties exposed to tropical-storm-force winds for \$55.0 million (34%), counties exposed to major-hurricane-force winds for \$10.5 million (6%), and counties exposed to tropical-depression-force winds for \$1.6 million (1%), respectively. Tropical-storm-force and hurricane-force winds are more frequent than major-hurricane-force winds and also cover a wider area that affects many more counties, so they result in larger total damages.

On a per-event basis, the average losses are largest for the Hurricane category, at about \$113,700, followed by Major Hurricane exposure at roughly \$99,500 and Tropical Storm exposure at about \$10,100. This pattern points to a clear non-linearity in storm damages, consistent with the broader hurricane economics literature showing that damages rise more than proportionally with wind intensity (Nordhaus, 2006; Hsiang, 2010).

Losses are also highly concentrated geographically. Figure 4 shows that Louisiana accounts for by far the largest cumulative net revenue losses in the sample, at about \$85.0 million, followed by Florida (\$24.3 million), Texas (\$16.9 million), Alabama (\$16.5 million), and North Carolina (\$12.7

⁵We monetize the event-study estimates by removing the storm effect on the transformed scale, day by day, for each treated county–species-group observation inside the estimated loss window. For county i , species group s , and relative day τ in the loss window of exposure category k , the counterfactual revenue is $\tilde{Y}_{is\tau} = \sinh(\text{asinh}(Y_{is\tau}) - \hat{\beta}_{\tau k})$ and the net loss is $\tilde{Y}_{is\tau} - Y_{is\tau}$; losses outside the window are set to zero. The loss window for each category is the maximal run of consecutive post-exposure days whose upper 95% confidence bound lies below zero.

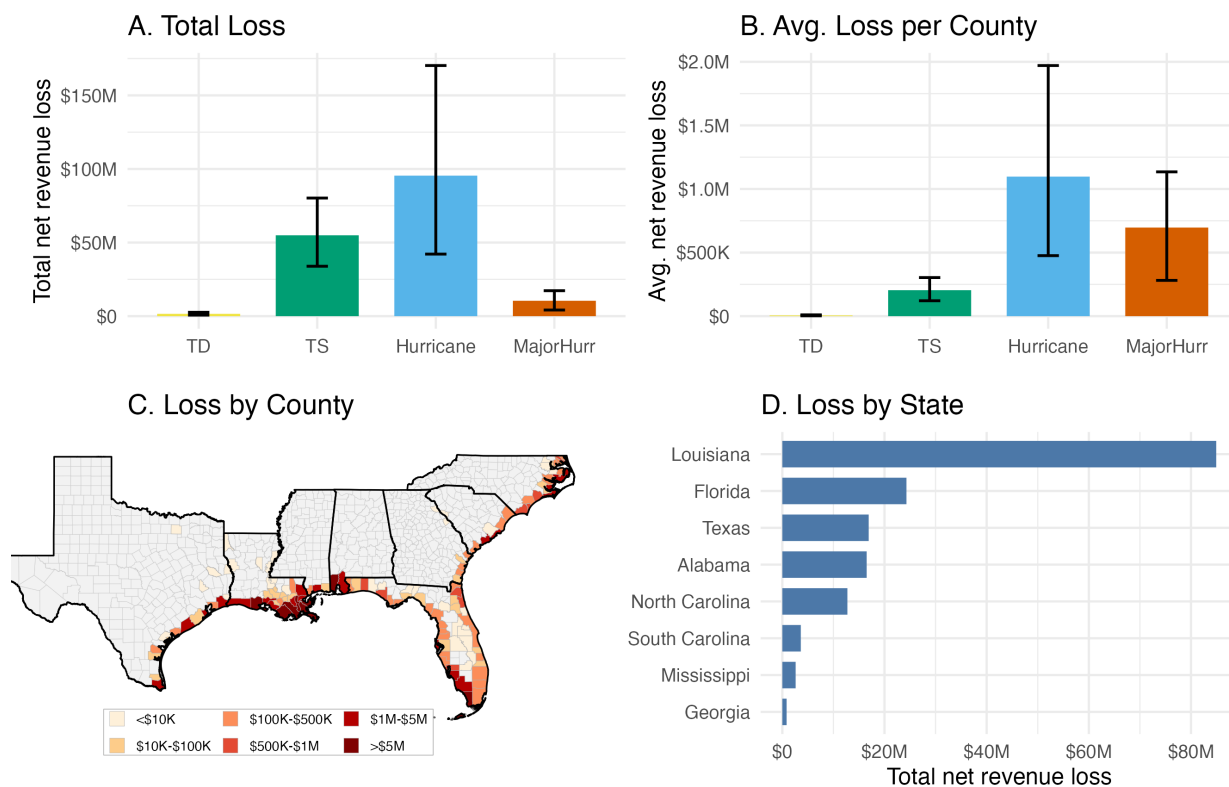


Figure 4: Cumulative revenue losses implied by the event-study estimates. Panel A reports total net revenue losses in 2022 dollars by county-level wind exposure category, obtained by summing the day-specific event-study effects over the estimated loss window for each category. Panel B reports the corresponding average cumulative loss per exposed county. Panel C maps cumulative net revenue losses across coastal counties in the Southeastern United States, summed across all exposure categories. Panel D reports total cumulative net revenue losses by state over 2005–2022. Exposure categories are Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s; Categories 1–2), and Major Hurricane (≥ 50 m/s; Category 3 or higher). Error bars in Panels A and B denote 95% bootstrap confidence intervals based on the county-level distribution of cumulative losses.

million).⁶

To properly scale our estimates, it is helpful to distinguish between acute, localized production shocks and broader, season-level revenue losses. While we document sharp daily revenue declines exceeding 70 percent under major-hurricane exposure, these disruptions are short-lived and geographically confined to a few counties per event. Monetizing these dynamic impacts implies \$162.4 million in total hurricane-induced revenue losses between 2005 and 2022, which averages to \$9.0 million per hurricane season, or about 1.8 percent of average hurricane-season (June 1–November 30) ex-vessel revenue across our sample counties.

This regional share is modest because exposure, while common, is uneven and the disruption it causes is brief. In a typical season roughly two-thirds of sample counties experience some storm-wind exposure, and a county that is exposed is hit about 1.4 times on average that season. Conditional on exposure, however, the effect is large: within exposed counties, revenue falls roughly one third below its no-hurricane counterfactual over the disruption window.

Losses are much larger for especially severe seasons with repeated hurricanes. In 2020, for example, losses reached \$71.5 million, or roughly 13 percent of that year’s hurricane-season ex-vessel revenue in our sample counties. Taken together, these results suggest that hurricanes generate intense but temporary disruptions to fisheries production, with aggregate effects concentrated in active storm seasons rather than persistent contractions in regional production.

A natural question is whether these losses represent genuine foregone income or a redistribution of activity over time, particularly for species managed under individual fishing quotas (Drymon et al., 2023), where a fisher who misses trips during a storm may recoup catch later in the season. Our design does not net out such within-season substitution, and to the extent it occurs, our estimates would overstate the change in annual income. We argue that two features temper this concern. First, the disruption is prolonged rather than instantaneous. Landings and revenue remain depressed for two to four weeks under hurricane- and major-hurricane-force exposure, which shrinks the remaining season over which lost effort can be recovered, especially for late-season storms.

⁶The headline total is not an artifact of the Gulf Shrimp System duplication (which affects crustacean landings before 2016) or of geographic and temporal concentration. Holding the event-study coefficients fixed and re-aggregating the monetized losses, restricting the sample to the post-duplication period (2016–2022) implies an annualized loss of \$16.8 million per year, above the \$9.0 million full-sample average, indicating that the earlier, potentially duplicated years do not inflate the estimate. Excluding Louisiana, which accounts for roughly half of the cumulative total, leaves \$77.5 million, and excluding the severe 2020 season leaves \$91.0 million.

Second, the fisheries that account for the bulk of our losses are not managed under individual quotas: catch-share (IFQ) species make up only about 5% of the cumulative total, whereas non-catch-share fisheries, led by shrimp at roughly three-quarters, account for the remaining 95%, so their foregone harvest is not mechanically recoverable. We nonetheless view a formal decomposition of transitory versus permanent losses, for example by comparing daily effects against annual landings totals, as a valuable extension, and we interpret our estimates as short-run production losses rather than net changes in annual income. We note that foregone revenue overstates the net profit loss to fishers, since vessels that remain in port avoid the variable costs of fishing such as fuel, bait, ice, and crew shares. This production-side upper bound is distinct from, and does not contradict, our characterization of the estimates as a lower bound on the total economic burden, which omits capital destruction and downstream spillovers.

These estimates also help reconcile differences with post-disaster assessments, such as those produced by NOAA Fisheries, which often report substantially larger economic impacts for individual storms (National Oceanic and Atmospheric Administration, 2018b,a,c; Caffey et al., 2022). Those assessments typically combine multiple margins, including capital destruction (e.g., vessels, docks, and processing facilities), downstream supply-chain disruptions, and, in some cases, recreational fishing losses and input-output spillovers. In contrast, our framework isolates forgone revenue from commercial harvesting alone. Accordingly, our estimates should not be interpreted as capturing the total economic loss of hurricanes to fishing communities, but rather the commercial production-side component of that loss. This distinction is important because, while fishing operations are highly sensitive to short-term weather shocks, they also exhibit considerable resilience, with output recovering once conditions normalize.

4.3 Policy Implications and Disaster Preparedness

The results suggest that disaster policy for fisheries should remain focused on catastrophic landfalls for their physical destruction, but it should also become more proactive regarding the broader set of localized wind exposures that repeatedly disrupt fisheries production across a wide coastal footprint. Although catastrophic storms cause severe physical damage and generate the sharpest immediate disruptions, recurrent lower-magnitude exposures of many more counties to hurricane-force and tropical-storm-force winds drive the majority of cumulative revenue losses. This finding has a

direct policy implication: preparedness and relief eligibility triggered only by catastrophic-storm thresholds misses the bulk of the realized burden. Relief frameworks should extend to the recurrent tropical-storm- and hurricane-force exposures that generate most of the economic burden on commercial fishing communities.

The timing of the losses also matters for policy design. Effective interventions are likely those that shorten the disruption window rather than those that attempt to eliminate all losses *ex ante*. Because production remains depressed for two to three weeks under hurricane-force exposure and up to a month under major-hurricane-force exposure, cumulative losses accrue over the recovery window rather than on impact alone. Rapid restoration of basic operational capacity—docks, ice, fuel, power, and port access—is therefore central to limiting cumulative impacts, and shaving even a few days off restoration could avert a disproportionate share of the loss. Evidence shows that hurricane-impacted seafood systems restart only after electricity and other cold-chain infrastructure are restored, with recovery mirroring the pace of power restoration (Agar et al., 2020). This is true for regions where most catch is absorbed by supply chains fulfilling the demand of areas not impacted by hurricanes (i.e., shrimp caught in the Gulf but consumed in California); smaller seafood sectors that are heavily reliant on local demand may be slower to recover. This pattern aligns with our finding that the duration of disruption, rather than the magnitude of the initial shock, drives cumulative losses. Our state-level results likewise point to the value of geographically targeted investments in enhancing resilience, especially in Louisiana, Florida, Texas, Alabama, and North Carolina, which account for the largest cumulative losses in the sample.

The estimates also speak directly to fishery disaster policy under NOAA’s current framework. NOAA evaluates fishery resource disasters using sector-specific revenue losses, with losses above 80 percent potentially supporting a positive determination and losses between 35 and 80 percent subject to further evaluation (NOAA Fisheries, 2025). Our results show why a dynamic causal approach is valuable in that context. In practice, this suggests that fisheries disaster preparedness should combine better early warning communication with faster post-storm restoration and a more explicit recognition that repeated, spatially extensive wind exposure can generate large cumulative losses over time even when the underlying storms are not the most extreme in the sample.

Finally, our estimates should be interpreted as a small portion of the total economic burden of hurricanes on fishing communities. They isolate revenue losses associated with production disruption

tions, but do not include damage to vessels, gear, docks, marinas, dealer and processor facilities, or broader spillovers to processing, distribution, and local labor markets. This distinction matters when comparing our results to the disaster-assessment literature. For instance, NOAA’s evaluation of Hurricanes Irma and Maria reported \$106 million in revenue losses and an additional \$117 million in physical damages across Florida, Puerto Rico, and the U.S. Virgin Islands (National Oceanic and Atmospheric Administration, 2018c). Agar et al. (2020) similarly estimate \$17.8 million in losses for Puerto Rico’s small-scale fisheries after Hurricane Maria, of which 51% reflected damaged capital and shoreside infrastructure and 49% forgone fishing revenue. Likewise, a Louisiana Sea Grant and LDWF assessment of Hurricanes Laura, Delta, Zeta, and Ida (2020-21 seasons) estimated roughly \$579 million in total fisheries losses, including about \$305 million in infrastructure damage, \$155 million in revenue losses, and \$118 million in resource losses (Caffey et al., 2022). Against this context, our estimates suggest that the total economic burden of hurricanes in the sector is likely on the order of two to four times larger, and concentrated further down the value chain. These ratios are consistent with previous work, which has shown that the net contribution of fisheries to the global economy is around 2.8 times larger than the ex-vessel value of catch (Dyck and Sumaila, 2010).

The central contribution of this work is to isolate the production margins through which hurricanes depress commercial fisheries production. Contextualizing our estimates against existing post-disaster assessments clarifies the specific margins of vulnerability within the commercial fishing sector. Our framework estimates \$162.4 million in cumulative revenue losses between 2005 and 2022, capturing a localized penalty of operational downtime. Crucially, the temporal dynamics in our models show that production remains depressed well beyond the period of hazardous weather itself. This shows that, while dangerous conditions typically subside within a few days of exposure, landings and revenue remain significantly below baseline for two to three weeks under hurricane-force exposure and up to a month under major-hurricane-force exposure. This persistent gap suggests that the mechanisms depressing production extend beyond unsafe seas: damage to vessels, gear, docks, fishers’ homes and processing infrastructure constrain operational capacity, while ecological disruptions to nearshore habitats may reduce the availability of target species even after fishers are able to return to sea. Damage to fishers’ own homes may be a first-order driver of this prolonged recovery rather than a peripheral factor. Like other residents, fishers prioritize

the safety of their families and the structural integrity of their homes before resuming operations, so households facing displacement or uninhabitable dwellings may stay out of the fishery well after seas calm and shoreside infrastructure is restored. This household-level channel would be consistent with the extended recovery paths we observe under the most severe exposure categories. Operational disruption, capital destruction, and ecological degradation likely operate sequentially, with hazardous weather driving the immediate shock and capital and ecological damage prolonging the recovery.

5 Conclusion

We provide evidence on the economic impacts of hurricanes on commercial fisheries in the Southeast United States. Using a high-frequency county-level panel that links daily landings and revenue to realized local wind exposure, we show that hurricane-related wind exposure leads to large and persistent reductions in fisheries production. The effects are strongly non-linear: counties exposed to stronger local winds experience larger and more persistent declines in both landings and revenue. Yet cumulative losses are not driven solely by extreme exposures; more frequent tropical-storm- and hurricane-force winds, which affect many more counties, account for much of the total economic burden.

Our findings also highlight the importance of timing. Fishers reduce production in advance of exposure, experience the largest disruptions around impact, and then recover only gradually over the following days and weeks. These dynamic patterns matter for measurement. Simple non-causal comparisons understate the economic costs of hurricanes and can even deliver misleading conclusions, whereas the event-study framework captures anticipatory behavior, contemporaneous disruption, and recovery within a unified causal design. Summing these dynamic effects implies sizable cumulative revenue losses for the Southeast U.S. commercial fishing sector over 2005–2022, with losses concentrated in a relatively small set of highly exposed coastal states.

From a policy standpoint, the results suggest that targeted investments in preparedness, infrastructure resilience, and rapid recovery can yield meaningful returns. The most effective interventions are likely those that reduce the duration of operational disruption by restoring power, fuel, ice, port access, and market logistics as quickly as possible after exposure (Villaseñor-Derbez et al.,

2026). More broadly, the findings indicate that fisheries disaster policy should not focus exclusively on the rarest catastrophic storms, but also on the repeated, spatially extensive wind exposures that generate substantial cumulative losses over time. Because the revenue losses estimated here exclude capital damage and broader downstream spillovers, they should be interpreted as a lower bound on the total economic burden faced by fishing communities.

Future research could build on these results by examining distributional consequences across fisheries and regions, the interaction between disaster assistance and private adaptation, and the extent to which changing storm patterns may alter the long-run viability of vulnerable fishing systems. By quantifying the direct production losses associated with hurricane exposure in a causal framework, this study provides a benchmark for evaluating disaster policy and for designing more resilient coastal economies under a changing climate.

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Appendix

Supplementary Tables and Figures

For

*The Economic Impacts of Hurricanes on Fisheries
in the United States*

A Species-Level Composition of the Fishery

This appendix provides species-level detail underlying the broader group-level patterns reported in the main text. Table A1 reports the 20 species with the highest average annual revenue in the sample, while Table A2 reports the 20 species with the highest average annual landings.

The revenue rankings show that the economic value of the fishery is concentrated in a relatively small number of species. Northern white shrimp, northern brown shrimp, blue crab, eastern oyster, and menhadens account for a large share of total revenue, highlighting the outsized commercial importance of crustaceans and mollusks in particular. By contrast, the landings rankings are even more concentrated in high-volume species, led overwhelmingly by menhadens, followed by bait shrimp, northern white shrimp, blue Crab, and northern brown shrimp. These comparisons illustrate the meaningful differences between value concentration and volume concentration across the fishery.

Table A1: Top 20 Species by Average Annual Revenue (2005–2022)

Species	Group	Avg. Annual Revenue	Avg. Price (\$/kg)	Revenue Share
Shrimp, Northern White	Crustaceans	\$192.4M	\$ 4.55	22.7%
Shrimp, Northern Brown	Crustaceans	\$118.4M	\$ 4.22	14.0%
Crab, Blue	Crustaceans	\$100.4M	\$ 2.38	11.9%
Oyster, Eastern	Mollusks	\$ 77.7M	\$ 4.01	9.2%
Menhadens	Pelagic Fish	\$ 65.0M	\$ 0.19	7.7%
Lobster, Caribbean Spiny	Crustaceans	\$ 35.2M	\$16.12	4.2%
Crab, Stone	Crustaceans	\$ 29.0M	\$25.57	3.4%
Shrimp, Northern Pink	Crustaceans	\$ 24.2M	\$ 5.39	2.9%
Snapper, Red	Reef Fish	\$ 19.0M	\$ 9.75	2.2%
Grouper, Red	Reef Fish	\$ 14.8M	\$ 7.99	1.7%
Procambarus	Other Species	\$ 11.5M	\$ 2.19	1.4%
Mackerel, King	Pelagic Fish	\$ 11.2M	\$ 4.56	1.3%
Flounders, Paralichthys	Other Species	\$ 10.3M	\$ 5.72	1.2%
Mullet, Striped	Inshore Fish	\$ 8.8M	\$ 1.51	1.0%
Shrimps, Bait	Crustaceans	\$ 8.7M	\$ 0.19	1.0%
Tuna, Yellowfin	Pelagic Fish	\$ 7.6M	\$ 9.12	0.9%
Swordfish	Pelagic Fish	\$ 7.3M	\$ 9.39	0.9%
Snapper, Vermilion	Reef Fish	\$ 7.2M	\$ 6.88	0.9%
Snapper, Yellowtail	Reef Fish	\$ 5.5M	\$ 7.32	0.7%
Grouper, Gag	Reef Fish	\$ 5.2M	\$10.24	0.6%

Note:

Species ranked by average annual revenue across all counties and years (2005–2022). Revenue share is each species' share of total sample revenue. Common names are as recorded in NOAA dealer reports.

Table A2: Top 20 Species by Average Annual Landings (2005–2022)

Species	Group	Avg. Annual Landings (kg)	Avg. Price (\$/kg)	Landings Share
Menhadens	Pelagic Fish	344.5M	\$ 0.19	58.7%
Shrimps, Bait	Crustaceans	46.8M	\$ 0.19	8.0%
Shrimp, Northern White	Crustaceans	42.3M	\$ 4.55	7.2%
Crab, Blue	Crustaceans	42.2M	\$ 2.38	7.2%
Shrimp, Northern Brown	Crustaceans	28.1M	\$ 4.22	4.8%
Oyster, Eastern	Mollusks	19.4M	\$ 4.01	3.3%
Mullet, Striped	Inshore Fish	5.8M	\$ 1.51	1.0%
Procambarus	Other Species	5.3M	\$ 2.19	0.9%
Shrimp, Northern Pink	Crustaceans	4.5M	\$ 5.39	0.8%
Mackerel, King	Pelagic Fish	2.5M	\$ 4.56	0.4%
Lobster, Caribbean Spiny	Crustaceans	2.2M	\$16.12	0.4%
Mackerel, Spanish	Pelagic Fish	2.1M	\$ 2.15	0.4%
Snapper, Red	Reef Fish	1.9M	\$ 9.75	0.3%
Croaker, Atlantic	Inshore Fish	1.9M	\$ 1.47	0.3%
Grouper, Red	Reef Fish	1.9M	\$ 7.99	0.3%
Flounders, Paralichthys	Other Species	1.8M	\$ 5.72	0.3%
Drum, Black	Inshore Fish	1.7M	\$ 2.16	0.3%
Catfish, Blue	Other Species	1.6M	\$ 1.16	0.3%
Fishes, Buffalo	Other Species	1.4M	\$ 0.52	0.2%
Osteichthyes	Other Species	1.4M	\$ 0.78	0.2%

Note:

Species ranked by average annual landings in kilograms across all counties and years (2005–2022). Landings share is each species' share of total sample landings. Common names are as recorded in NOAA dealer reports.

A.1 Distribution of Exposure Events

Table A3 reports the distribution of the $\text{Post} \times \text{ExposureLev}$ variable across the four wind-exposure categories. Exposure is far more common at lower intensities: tropical-depression- and tropical-storm-force events each number over five thousand, hurricane-force events number 840, and major-hurricane-force events only 105 (in 15 counties). This steep frequency gradient—tropical-storm-force exposure occurs roughly six to seven times as often as hurricane-force exposure—is why the more frequent, moderate exposures account for most of the cumulative economic burden despite their smaller per-event effect.

Table A3: Distribution of exposure events and observations by wind-exposure category

Exposure Level	Events	Counties	Pre-Period Obs.	Post-Period Obs.	Total Obs.
Tropical Depression	5,138	260	154,140	186,844	340,984
Tropical Storm	5,432	269	162,960	255,290	418,250
Hurricane	840	87	25,200	45,045	70,245
Major Hurricane	105	15	3,150	5,026	8,176
Total	11,515	275	345,450	492,205	837,655

Note:

The table reports the distribution of the $\text{Post-by-ExposureLev}$ interaction across the four wind-exposure categories. An event is a county-level exposure episode, formed by grouping a county's hazardous-wind days within 30 days of one another and classified by the maximum wind intensity reached (Tropical Depression, 10–17.5 m/s; Tropical Storm, 17.5–33 m/s; Hurricane, 33–50 m/s; Major Hurricane, 50 m/s or more). Observations are county-by-date-by-species-group cells in the 30-day pre- and post-exposure windows around each event. County counts do not sum across categories because a county may experience events of different intensities over the sample.

B Static Analysis

B.1 Static Effects Controlling for Precipitation Bins

This appendix evaluates whether the static pre-post effects attributed to county-level wind exposure are sensitive to controlling flexibly for contemporaneous precipitation. To do so, we re-estimate the main 30-day pre-post specification after adding controls for observed precipitation bins. Specifically, observed precipitation is winsorized at 200 mm and entered as a set of 20 mm bins, with the lowest bin serving as the omitted category. This allows the wind-exposure coefficients to be identified net of non-linear variation in rainfall exposure.

Figure B1 shows that the main static results are largely unchanged. The estimated effects remain close to zero for tropical depressions, negative and moderate for tropical storms, and much larger for hurricane-force and major-hurricane-force exposure. Controlling flexibly for precipitation therefore attenuates the coefficients only modestly and does not alter the main non-linear pattern documented in the text.

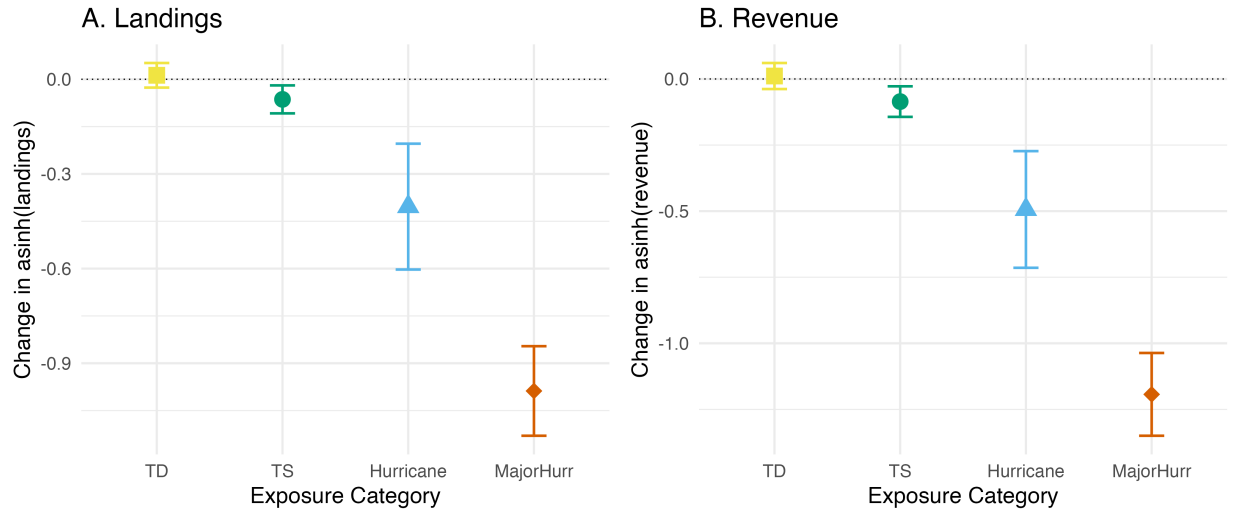


Figure B1: Pre-post effects of hurricane exposure on commercial fisheries by exposure level, controlling for observed precipitation bins. Panel A shows the estimated change in the inverse hyperbolic sine of daily landings, while Panel B shows the corresponding change in the inverse hyperbolic sine of daily revenue. Each coefficient represents the difference between the 30-day post-exposure and 30-day pre-exposure windows around a county-level exposure event for the indicated wind exposure level. The specification includes the same fixed effects as the main pre-post model, along with observed precipitation-bin controls based on 20 mm bins of winsorized precipitation. Bars represent 95% confidence intervals based on Conley standard errors.

B.2 Static Effects by Species Group

This appendix presents the static pre-post estimates separately by species group. Figures B2 and B3 re-estimate the main 30-day pre-post specification for each of the seven broad species groups in the data. This breakdown shows that the aggregate static effects reported in the main text are broadly shared across the fishery, but also reveals meaningful heterogeneity in the scale of the disruption across groups.

Several patterns stand out. First, crustaceans are the most severely affected group by a substantial margin. Their coefficients are negative and statistically precise across all exposure categories above tropical depression, with especially large declines under hurricane-force and major-hurricane-force exposure in both landings and revenue. Second, other species and inshore fish also exhibit large negative responses, particularly under hurricane and major-hurricane exposure, while mollusks and pelagic fish show clear but more moderate contractions. Reef fish are also negatively affected, though their responses to tropical-storm exposure are smaller and less precisely estimated. By contrast, sharks and rays display the weakest and noisiest responses in the sample, with relatively small coefficients and little evidence of systematic major-hurricane effects.

These results reinforce the interpretation that the aggregate non-linearity documented in the main text is not driven by a single sector alone. Rather, stronger local wind exposure generates larger static losses across most fisheries, although the burden is concentrated most heavily in crustaceans and a subset of other nearshore species groups.

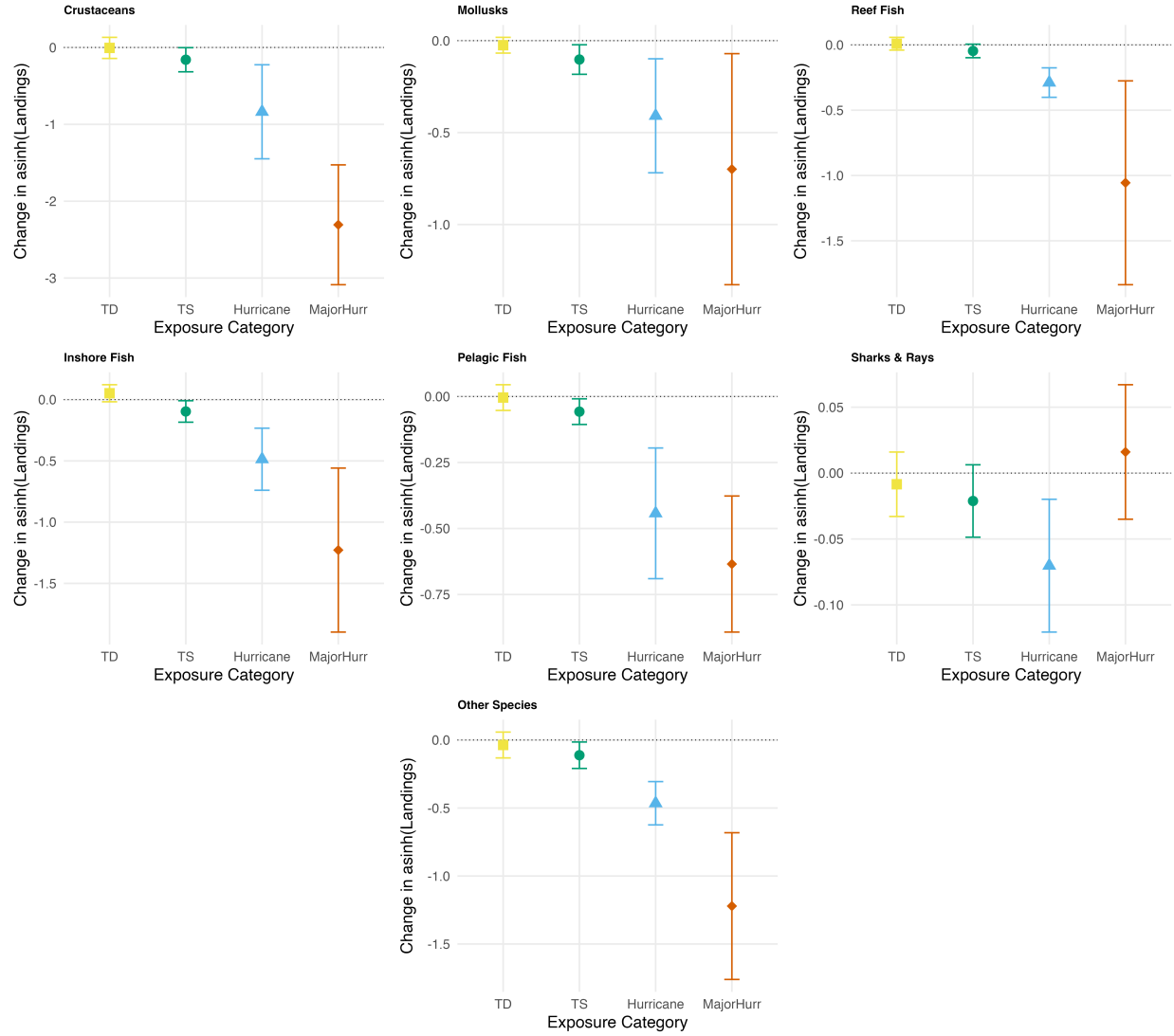


Figure B2: Pre-post effects on landings by species group and county-level wind exposure category. Each panel corresponds to one species group. Within each panel, points represent coefficient estimates from the 30-day pre-post specification for Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), and Major Hurricane (≥ 50 m/s), and bars denote 95% confidence intervals based on Conley standard errors. The dependent variable is the inverse hyperbolic sine of daily landings. All regressions include county-by-month, year, and day-of-week fixed effects.

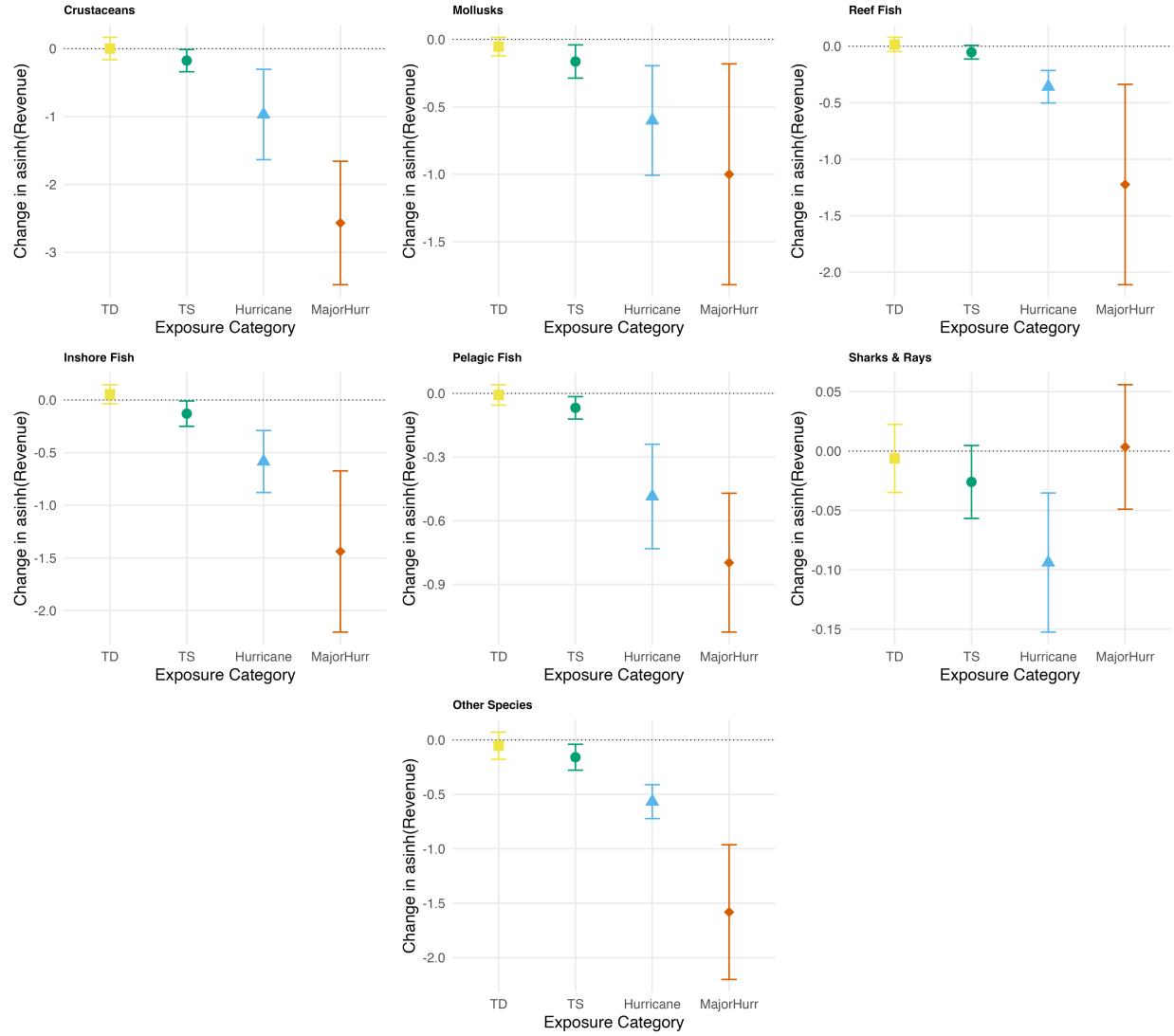


Figure B3: Pre-post effects on revenue by species group and county-level wind exposure category. Each panel corresponds to one species group. Within each panel, points represent coefficient estimates from the 30-day pre-post specification for Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), and Major Hurricane (≥ 50 m/s), and bars denote 95% confidence intervals based on Conley standard errors. The dependent variable is the inverse hyperbolic sine of daily revenue. All regressions include county-by-month, year, and day-of-week fixed effects.

B.3 Sensitivity to the Event-Definition Window

Our baseline analysis groups county-level exposure days within 30 days into a single event and estimates effects over a symmetric 30-day pre/post window. We assess whether this choice drives our findings by re-estimating the static pre-post specification using 14- and 21-day event windows. Narrower windows cluster exposure days more tightly, which addresses the concern that a 30-day window may mask overlapping or repeated shocks in active hurricane seasons.

Table B1 reports the results. The non-linear gradient in effects is robust across all three windows: estimated declines are near zero for tropical-depression exposure, moderate for tropical-storm exposure, and substantially larger for hurricane-force and major-hurricane-force exposure, in both landings and revenue. Point estimates are larger at shorter windows because a narrower window concentrates on the days of most acute disruption, whereas the 30-day window averages in the subsequent recovery. The 30-day specification reported in the main text is therefore the most conservative of the three. Overall, the qualitative conclusion that daily landings and revenue decline sharply and non-linearly with local wind intensity does not depend on the event-definition window.

Table B1: Sensitivity of pre-post estimates to the event-definition window

Exposure Level	asinh(Landings)			asinh(Revenue)		
	14-day	21-day	30-day	14-day	21-day	30-day
Tropical Depression	-0.042*** (0.014)	-0.022 (0.014)	-0.003 (0.019)	-0.057*** (0.018)	-0.028 (0.018)	-0.007 (0.025)
Tropical Storm	-0.183*** (0.047)	-0.117*** (0.040)	-0.085*** (0.023)	-0.225*** (0.056)	-0.143*** (0.049)	-0.111*** (0.031)
Hurricane	-0.599*** (0.093)	-0.507*** (0.106)	-0.428*** (0.105)	-0.728*** (0.108)	-0.612*** (0.117)	-0.523*** (0.116)
Major Hurricane	-1.395*** (0.105)	-1.233*** (0.105)	-1.019*** (0.076)	-1.689*** (0.112)	-1.490*** (0.117)	-1.229*** (0.085)
Observations	440,349	626,325	837,655	440,349	626,325	837,655

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Each cell reports the estimated change between the pre- and post-exposure windows for the indicated exposure level (coefficient on the interaction of Post and the exposure level), with Conley standard errors in parentheses. Columns vary the event-definition window used to group county-level exposure days into events and to define the symmetric pre/post estimation horizon (14, 21, or 30 days); the 30-day window is the baseline reported in the main text. Shorter windows concentrate on the days of most acute disruption and therefore yield larger point estimates. All specifications include county-by-month, year, day-of-week, and species-group fixed effects.

B.4 Static Effects in Levels

This appendix reports the static pre-post effects of county-level wind exposure in levels rather than using the inverse hyperbolic sine transformation. We re-estimate the main 30-day pre-post specification with daily landings and daily revenue in levels, preserving the same event construction, exposure categories, and fixed effects as in the main text. Because levels specifications place greater weight on a relatively small number of high-volume and high-value observations and retain the large mass at zero in the underlying outcomes, they are mechanically noisier than the corresponding *asinh* estimates. For that reason, they are less useful for precise inference about average magnitudes, but they remain informative about sign and relative ordering across exposure categories.

Figure B4 shows that the same broad non-linearity remains visible in levels. The estimated effects are comparatively small for tropical-depression exposure, larger and more negative for tropical-storm exposure, and substantially larger for hurricane-force and major-hurricane-force exposure, especially for revenue. Thus, while the levels specification is noisier than the main specification, it supports the same basic conclusion that stronger local wind exposure is associated with larger reductions in fisheries production.

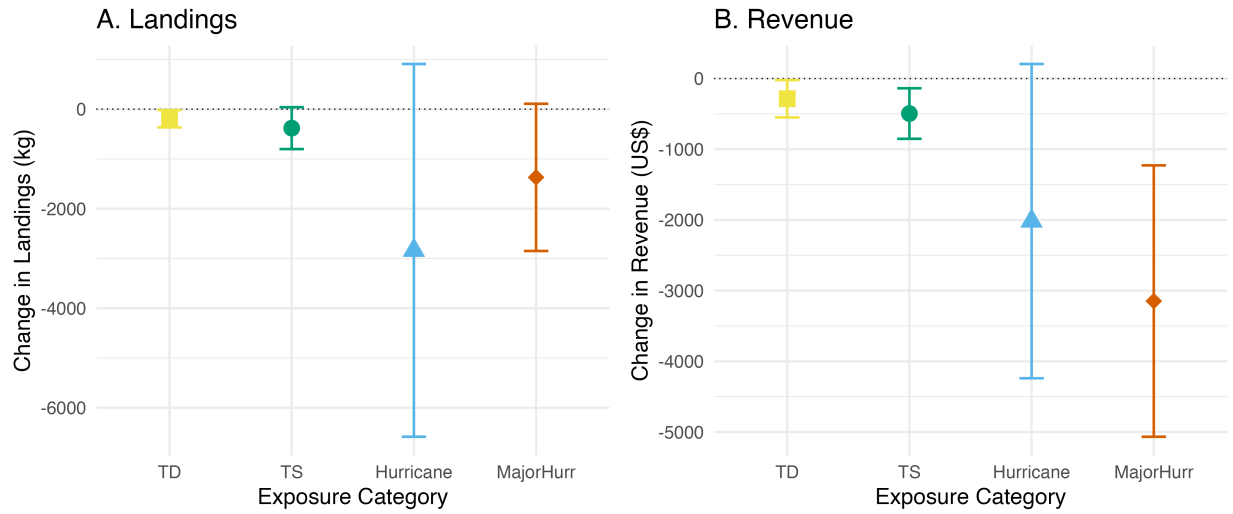


Figure B4: Pre-post effects of hurricane exposure on commercial fisheries by exposure level in levels. Panel A shows the estimated change in daily landings, while Panel B shows the corresponding change in daily revenue. Each coefficient represents the difference between the 30-day post-exposure and 30-day pre-exposure windows around a county-level exposure event for the indicated wind exposure level. Levels are Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s; Categories 1–2), and Major Hurricane (≥ 50 m/s; Category 3 or higher). Bars represent 95% confidence intervals based on Conley standard errors. The regressions include county-by-month, year, day-of-week, and species-group fixed effects.

B.5 Robustness to the Fisheries Data Limitation

As noted in Section 3, the fisheries data contain duplicate shrimp records in the Gulf Shrimp System over 2005–2015 that may inflate reported crustacean landings. Because crustaceans account for the largest share of revenue and the largest estimated effects, we assess whether this limitation drives our results. Table B2 re-estimates the 30-day pre-post specification excluding the crustaceans group, excluding Gulf states, and restricting the sample to 2016–2022, after the duplicate-record period.

The results are essentially unchanged across all specifications. Excluding crustaceans or Gulf states leaves the non-linear gradient intact, with only modest attenuation. Restricting the sample to 2016–2022 yields effects that are, if anything, slightly larger than the baseline. We take this as evidence that the measured effects are not an artifact of the data limitation.

Table B2: Robustness of pre-post estimates to the fisheries data limitation

Exposure	Baseline	Drop Crustaceans	Drop Gulf States	2016–2022
Panel A: asinh(Landings)				
Tropical Depression	-0.003 (0.019)	-0.002 (0.013)	-0.010 (0.016)	-0.019 (0.016)
Tropical Storm	-0.085*** (0.023)	-0.073*** (0.019)	-0.068*** (0.017)	-0.082*** (0.018)
Hurricane	-0.428*** (0.105)	-0.360*** (0.076)	-0.316** (0.133)	-0.434*** (0.099)
Major Hurricane	-1.019*** (0.076)	-0.804*** (0.106)	-1.076*** (0.055)	-1.187*** (0.092)
Panel B: asinh(Revenue)				
Tropical Depression	-0.007 (0.025)	-0.008 (0.017)	-0.016 (0.019)	-0.027 (0.020)
Tropical Storm	-0.111*** (0.031)	-0.100*** (0.026)	-0.080*** (0.024)	-0.099*** (0.024)
Hurricane	-0.523*** (0.116)	-0.448*** (0.086)	-0.426*** (0.161)	-0.539*** (0.115)
Major Hurricane	-1.229*** (0.085)	-1.006*** (0.117)	-1.275*** (0.107)	-1.438*** (0.105)
Observations	837,655	717,990	492,849	598,458

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Each cell reports the coefficient on the interaction of Post and the exposure level from the 30-day pre-post specification, with Conley standard errors in parentheses. Columns re-estimate the model (i) on the full sample; (ii) excluding the crustaceans group, which contains the shrimp species subject to duplicate records in the Gulf Shrimp System over 2005–2015; (iii) excluding Gulf states (Texas, Louisiana, Mississippi, and Alabama); and (iv) restricting to 2016–2022, after the duplicate-record period. All specifications include county-by-month, year, day-of-week, and species-group fixed effects.

B.6 Alternative Functional Forms and the asinh Interpretation

Our main results use the inverse hyperbolic sine (asinh) transformation to accommodate the mass of zeros in daily county-level activity. Because asinh coefficients are not exact percentage changes and can depend on the scale of the outcome, we benchmark the implied magnitudes against three alternatives. Table B3 reports the implied percentage change under each.

For revenue, the approaches agree closely. Poisson pseudo-maximum-likelihood (PPML) requires no transformation of zeros and provides an exact semi-elasticity. This implies revenue declines of 44 and 80 percent under hurricane- and major-hurricane-force exposure, respectively, and closely match the asinh-based figures of 41 and 71 percent in our main specification. A $\log(y + 1)$ transformation, which retains the zeros through a unit shift, yields a similar picture (38 and 67 percent). Together these suggest our percentage interpretation in the main text is reasonable and not an artifact of the asinh transformation.

For landings, the estimates are more method-dependent due to the variable's right-skewed distribution. PPML implies larger declines than asinh (66 versus 35 percent under hurricane-force exposure), whereas the $\log(y + 1)$ transformation closely tracks asinh (32 percent). This indicates that the divergence reflects PPML's sensitivity to the conditional mean of a heavily skewed outcome rather than the asinh transformation itself. The estimators nonetheless agree on sign, significance, and ordering across exposure levels. We retain the asinh estimates in the main text for consistency, but the PPML results suggest our reported magnitudes for landings might be conservative.

Finally, regressing the log outcome on the subsample with positive activity isolates the intensive margin. These estimates are markedly smaller, showing only a 20 percent revenue decline under hurricane-force exposure. The gap between these and the total-effect estimates highlights the extensive margin: roughly half of the total revenue decline arises from county-days that cease activity entirely, rather than from smaller catches on active days.

Table B3: Benchmarking the asinh magnitude interpretation: implied percentage change by specification

Exposure	Implied percent change			
	asinh	PPML	Log($y + 1$)	Log (drop zeros)
Panel A: Landings				
Tropical Depression	-0.3	-1.1	-0.3	-3.6
Tropical Storm	-8.2***	-16.7***	-7.3***	-4.4*
Hurricane	-34.8***	-66.0***	-32.0***	-22.0***
Major Hurricane	-63.9***	-76.3***	-59.8***	-29.4**
Observations	837,655	548,317	837,655	120,888
Panel B: Revenue				
Tropical Depression	-0.7	-12.9**	-0.7	-8.1**
Tropical Storm	-10.5***	-13.8**	-9.6***	-5.3*
Hurricane	-40.7***	-43.8***	-38.0***	-20.0***
Major Hurricane	-70.7***	-79.6***	-67.3***	-30.5***
Observations	837,655	548,317	837,655	122,022

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The asinh column is the inverse hyperbolic sine model reported in the main text. The PPML column is Poisson pseudo-maximum-likelihood. The $\log(y + 1)$ column adds one to the outcome before taking logs, retaining the zeros. The log (drop zeros) column regresses the log outcome on the subsample with positive activity. All specifications share the same fixed effects (county-by-month, year, day-of-week, species-group) and Conley standard errors. For PPML, fixed-effect groups with only zero outcomes are dropped.

C Dynamic Analysis

C.1 Dynamic Effects by Exposure Category

This appendix presents the event-study results separately by county-level wind exposure category. Figure C1 makes two patterns from the main text easier to see. First, the scale of the disruption rises sharply with local wind intensity. Tropical-depression effects are small and short-lived, tropical-storm effects are larger but still moderate, and the largest contemporaneous and cumulative disruptions occur under hurricane-force and major-hurricane-force exposure. Second, the timing of the response is also clearly non-linear: weaker exposure categories exhibit brief anticipatory declines and relatively rapid recovery, whereas hurricane-force and major-hurricane-force exposure generate deeper declines and substantially more persistent losses.

The separate category panels also clarify the role of anticipation and recovery. For tropical storms, fisheries production begins to contract shortly before exposure and remains depressed for roughly a week. For hurricane-force and major-hurricane-force exposure, the decline begins before day 0, intensifies sharply at impact, and recovers only gradually over the following weeks. Viewed this way, the appendix figure complements the combined main-text event-study plot by showing more directly that both the magnitude and persistence of the disruption increase with county-level wind exposure.

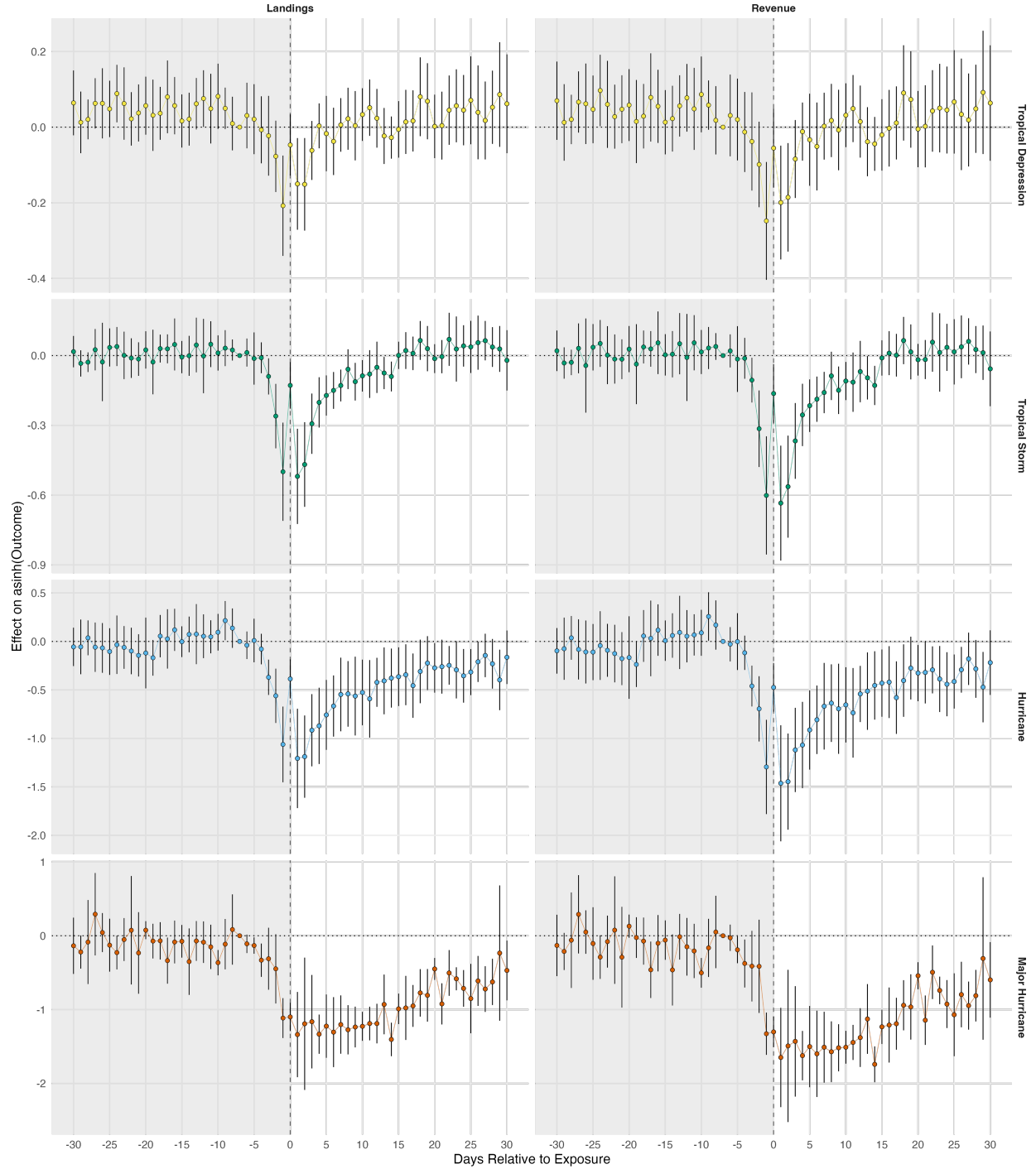


Figure C1: Dynamic effects of hurricane exposure by county-level wind exposure category. Each row corresponds to a different exposure category: Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), and Major Hurricane (≥ 50 m/s). Panel A reports the estimated effect on the inverse hyperbolic sine of daily landings, and Panel B reports the estimated effect on the inverse hyperbolic sine of daily revenue, for each day relative to county-level exposure (day 0). Points represent coefficient estimates, and bars denote 95% confidence intervals based on Conley standard errors. The regressions include county-by-month, year, day-of-week, and species-group fixed effects. The estimation sample contains 837,655 observations.

C.2 Dynamic Effects by Species Group

This appendix presents the dynamic event-study results separately by species group. Figures C2 and C3 show the same county-level exposure-specific event-study specification used in the main text, but estimated separately for each of the seven broad species groups in the data. This breakdown shows that the aggregate dynamics reported in the main text are not driven by a single fishery, but it also reveals meaningful heterogeneity in both the scale and persistence of the response.

Several patterns stand out. First, crustaceans are the most severely affected group. They exhibit the deepest declines in both landings and revenue, especially under hurricane-force and major-hurricane-force exposure, with large anticipatory contractions before day 0 and pronounced post-exposure losses that remain significant well beyond impact. Inshore fish and the residual “other species” group also show large and persistent responses, particularly under hurricane and major-hurricane exposure, with effects that frequently persist through much of the post-event window. By contrast, pelagic fish, mollusks, and reef fish still display clear negative responses, but these are generally smaller in magnitude than those for crustaceans and inshore fish, even when persistence remains substantial under stronger wind exposure. Sharks and rays are the least affected and noisiest group in the sample, with comparatively small and often statistically imprecise responses, especially for major-hurricane exposure.

Second, the timing patterns emphasized in the main text remain visible within species. For tropical storms, most groups begin to contract shortly before exposure and recover within roughly one to two weeks. For hurricane-force and major-hurricane-force exposure, however, the decline generally begins before day 0, deepens at or shortly after impact, and then fades only gradually over the following weeks. Taken together, these figures show that the main non-linearity documented in the aggregate results is present across a wide range of fisheries, but that the burden is particularly concentrated in crustaceans and a subset of other coastal and nearshore species groups.

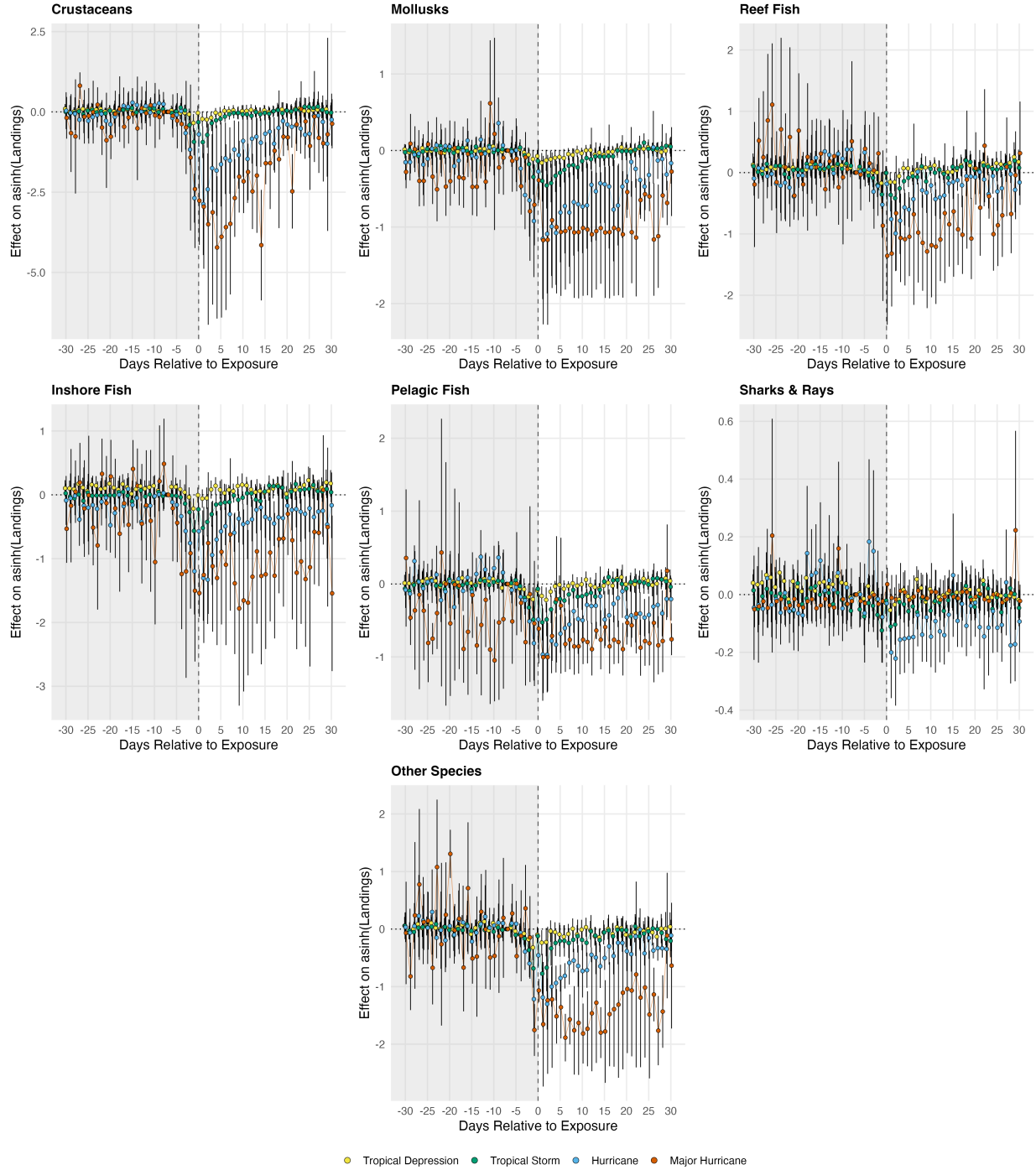


Figure C2: Dynamic effects on landings by species group and county-level wind exposure category. Each panel corresponds to one species group. Within each panel, points represent coefficient estimates from the event-study specification for Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), and Major Hurricane (≥ 50 m/s), and bars denote 95% confidence intervals based on Conley standard errors. The dependent variable is the inverse hyperbolic sine of daily landings, and coefficients are shown relative to day -7 . All regressions include county-by-month, year, and day-of-week fixed effects.

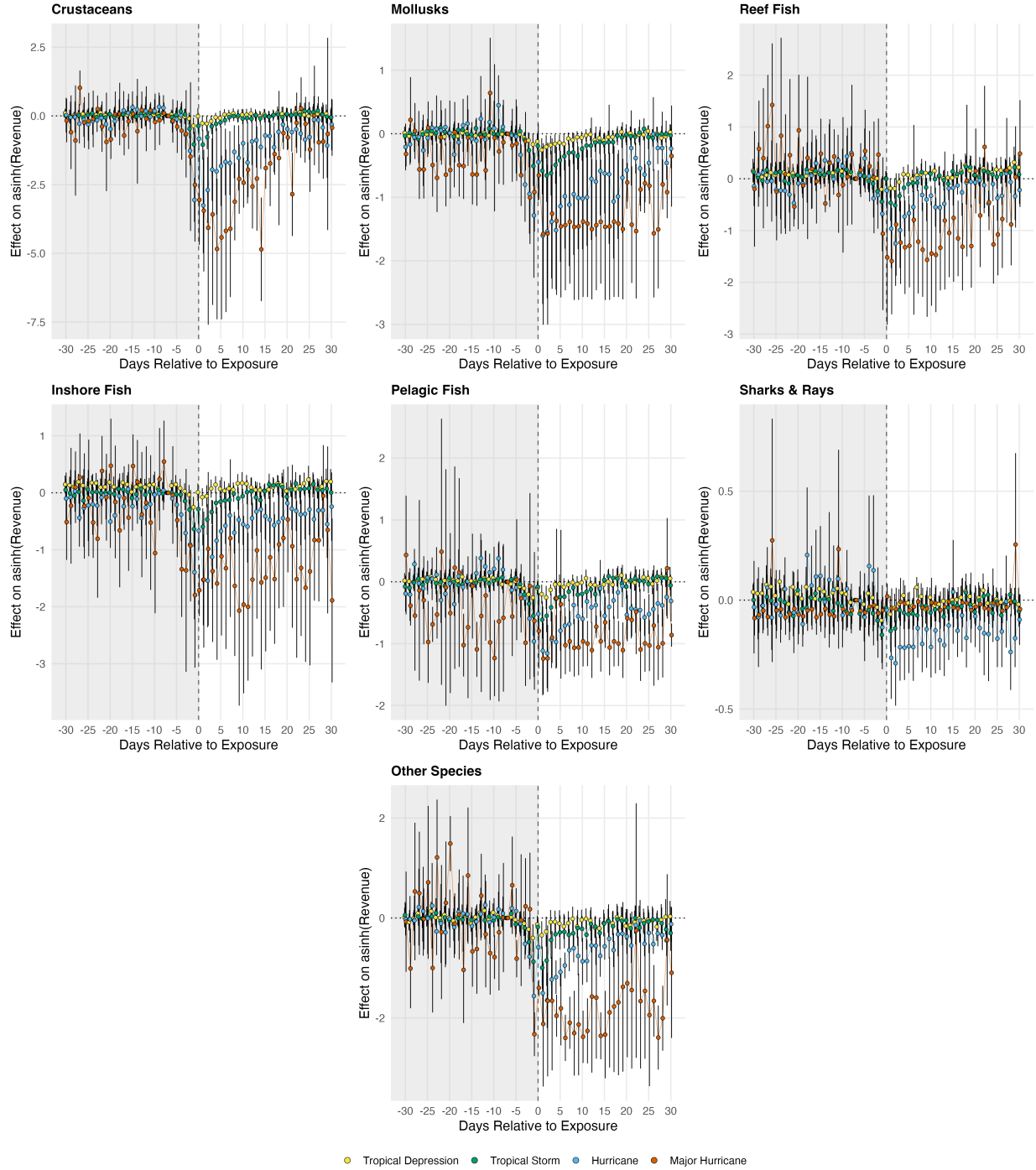


Figure C3: Dynamic effects on revenue by species group and county-level wind exposure category. Each panel corresponds to one species group. Within each panel, points represent coefficient estimates from the event-study specification for Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), and Major Hurricane (≥ 50 m/s), and bars denote 95% confidence intervals based on Conley standard errors. The dependent variable is the inverse hyperbolic sine of daily revenue, and coefficients are shown relative to day -7 . All regressions include county-by-month, year, and day-of-week fixed effects.

C.3 Dynamic Effects in Levels

This appendix reports the dynamic event-study results in levels rather than using the inverse hyperbolic sine transformation. The purpose is to assess whether the qualitative timing and non-linearity documented in the main text are sensitive to the functional-form choice. Because levels specifications place more weight on a relatively small number of high-value observations and retain the underlying mass at zero, they are mechanically noisier than the corresponding *asinh* specifications. For that reason, the levels estimates are less useful for precise inference about the exact day-by-day magnitude of the response.

Figure C4 nevertheless shows that the same broad patterns remain visible. The effects are small for tropical depressions, larger for tropical storms, and substantially larger for hurricane-force and major-hurricane-force exposure. The timing also remains similar: declines begin shortly before exposure, intensify around impact, and recover gradually afterward, with stronger wind exposure generating more persistent disruption. Thus, while the levels specification is noisier, it supports the same qualitative interpretation as the main *asinh*-based event study.

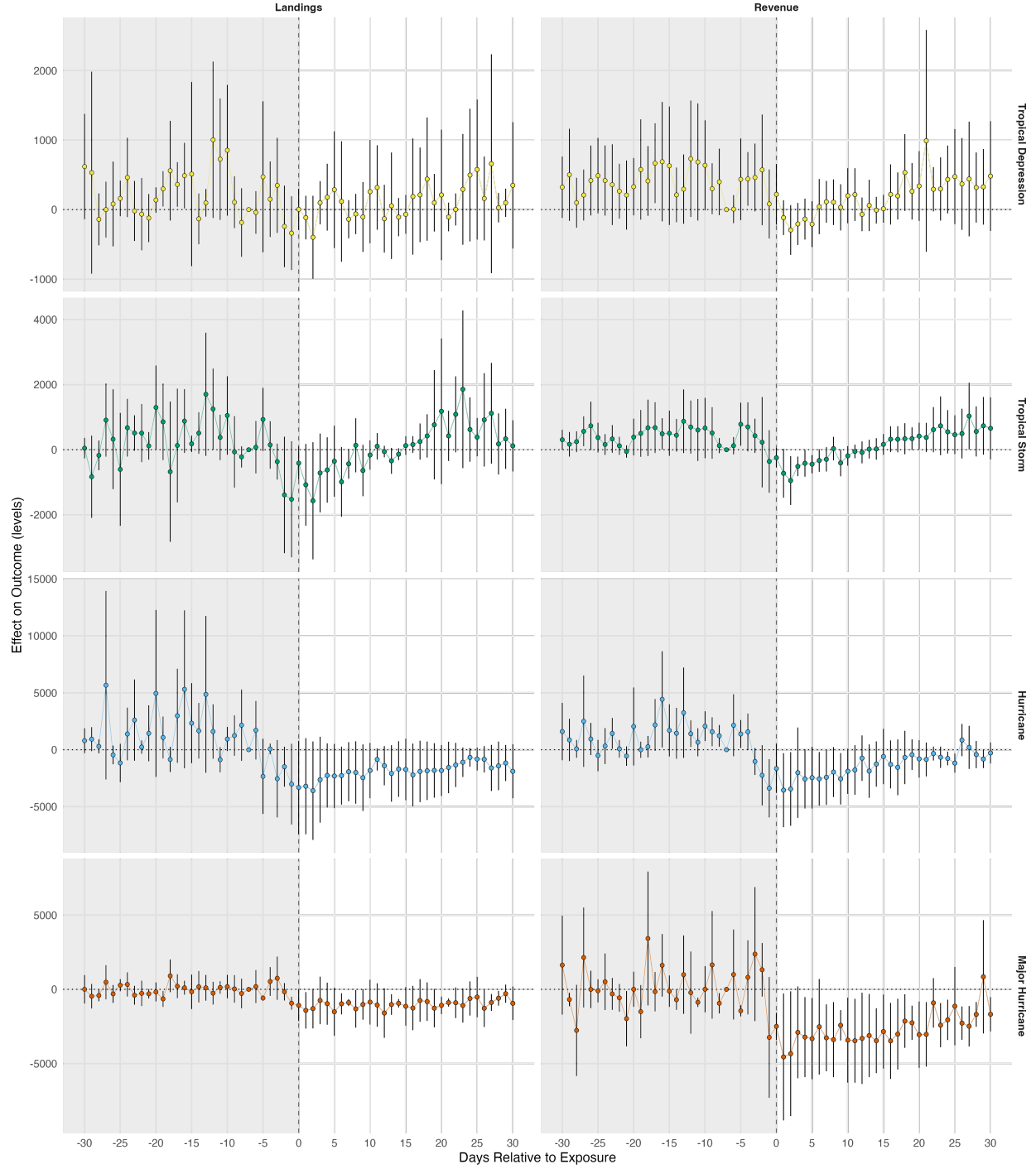


Figure C4: Dynamic effects of hurricane exposure by storm category in levels. Each row corresponds to a different exposure category: Tropical Depression (TD, 10–17.5 m/s), Tropical Storm (TS, 17.5–33 m/s), Hurricane (33–50 m/s), and Major Hurricane (≥ 50 m/s). The left column reports the estimated effect on daily landings, and the right column reports the estimated effect on daily revenue, for each day relative to county-level exposure (day 0). Points represent coefficient estimates, and bars denote 95% confidence intervals based on Conley standard errors. The regressions include county-by-month, year, day-of-week, and species-group fixed effects.

D Forecast Decomposition

This appendix decomposes the hurricane wind effect into an expected component, driven by forecast information available to fishers before exposure, and a surprise component arising from forecast errors. The analysis uses all county-day observations with positive observed wind speed and available forecast data (59,892 observations).

Table D1 presents the main results. Column (1) reports the baseline specification using realized wind speed only. Column (2) replaces realized wind with forecast wind and the forecast error (realized minus forecast), decomposing the total effect into what fishers could anticipate and the unexpected residual. The forecast component is roughly 2.6 times larger than the forecast error component, implying that approximately three-quarters of the total wind effect operates through the expected channel. This is consistent with fishers responding to forecast information by preemptively reducing activity, with the remaining quarter attributable to conditions that exceeded expectations. Column (3) confirms that forecast errors generate additional losses beyond what realized wind alone would predict. Column (4) shows that rapid intensification events, defined as episodes where realized wind exceeded the forecast by more than 10 m/s, generate disproportionately larger losses. Both panels show qualitatively similar patterns across revenue and landings.

Table D2 examines whether forecast errors matter differently across exposure categories by interacting the signed forecast error with exposure category indicators. The results reveal a striking asymmetry: forecast errors are insignificant for tropical storms and positive for tropical depressions, but large and negative for hurricane-force and major-hurricane-force exposure. This pattern implies that forecast surprises generate meaningful additional economic damage only at high wind intensities, where underprediction leaves fishers and infrastructure insufficiently prepared. At lower intensities, the operational margin for adjustment appears wide enough that forecast errors do not translate into measurable production losses. These patterns hold for both revenue and landings.

Table D3 disaggregates the forecast effect by lead time. The 1-day and 3-day forecasts are individually significant, while the 2-day forecast is not, likely reflecting collinearity among the three lead times. The forecast error remains significant across both landings and revenue, reinforcing the main decomposition result.

Table D1: Forecast Decomposition of Hurricane Impacts

	(1)	(2)	(3)	(4)
<i>Panel A: Revenue</i>				
asinh(Wind Speed)	−0.227*** (0.052)		−0.203*** (0.052)	−0.203*** (0.050)
Forecast Wind (m/s)		−0.046*** (0.009)		
Forecast Error (m/s)		−0.017*** (0.004)	−0.015*** (0.004)	
asinh(Wind Speed) × Rapid Intens.				−1.287*** (0.274)
<i>Panel B: Landings</i>				
asinh(Wind Speed)	−0.189*** (0.045)		−0.169*** (0.045)	−0.169*** (0.043)
Forecast Wind (m/s)		−0.038*** (0.007)		
Forecast Error (m/s)		−0.015*** (0.003)	−0.013*** (0.004)	
asinh(Wind Speed) × Rapid Intens.				−1.029*** (0.222)
Observations	59,892	59,892	59,892	59,892

Note:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are the inverse hyperbolic sine of daily revenue (Panel A) and landings in kg (Panel B). Column (1) regresses on realized wind speed only. Column (2) decomposes wind exposure into forecast wind (average 3-day forecast) and forecast error (realized minus forecast). Column (3) controls for both realized wind and the signed forecast error, where positive errors indicate underprediction (realized exceeded forecast). Column (4) interacts realized wind with a rapid intensification indicator (forecast error > 10 m/s). The rapid intensification main effect is included in the regression but not reported. All regressions include county-by-month, year, day-of-week, and species-group fixed effects. Standard errors are Conley spatial HAC.

Table D2: Forecast Error Effects by Exposure Category

	Signed Error	Error $\times$ Category
<i>Panel A: Revenue</i>		
asinh(Wind Speed)	−0.203*** (0.052)	−0.205*** (0.052)
Forecast Error (m/s)	−0.015*** (0.004)	
Error $\times$ Tropical Depression		0.033*** (0.011)
Error $\times$ Tropical Storm		−0.013 (0.008)
Error $\times$ Hurricane		−0.051*** (0.013)
Error $\times$ Major Hurricane		−0.039*** (0.008)
<i>Panel B: Landings</i>		
asinh(Wind Speed)	−0.169*** (0.045)	−0.170*** (0.046)
Forecast Error (m/s)	−0.013*** (0.004)	
Error $\times$ Tropical Depression		0.027*** (0.010)
Error $\times$ Tropical Storm		−0.013* (0.007)
Error $\times$ Hurricane		−0.042*** (0.010)
Error $\times$ Major Hurricane		−0.032*** (0.008)
Observations	59,892	59,892

Note:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are the inverse hyperbolic sine of daily revenue (Panel A) and landings in kg (Panel B). All columns control for realized wind speed. Column 1 adds the signed forecast error, where positive values indicate underprediction (realized wind exceeded the forecast). Column 2 interacts the signed forecast error with exposure category indicators. Forecast errors are calculated as observed minus average 3-day forecast wind speed. All regressions include county-by-month, year, day-of-week, and species-group fixed effects. Standard errors are Conley spatial HAC.

Table D3: Lead-Time Specific Forecast Effects

	Landings	Revenue
1-Day Forecast (m/s)	−0.025** (0.012)	−0.025* (0.015)
2-Day Forecast (m/s)	0.012 (0.014)	0.009 (0.018)
3-Day Forecast (m/s)	−0.030*** (0.011)	−0.036*** (0.013)
Forecast Error (m/s)	−0.019*** (0.004)	−0.024*** (0.004)
Observations	50,232	50,232

Note:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are the inverse hyperbolic sine of daily landings and revenue. Forecast wind speeds are entered separately by lead time (1-day, 2-day, and 3-day ahead) along with the forecast error (observed minus average forecast). The sample is restricted to observations with non-missing forecasts at all three lead times. All regressions include county-by-month, year, day-of-week, and species-group fixed effects. Standard errors are Conley spatial HAC.